



TECHNISCHE UNIVERSITÄT MÜNCHEN

Quantum Physics

Lucas Duckerschein



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1 Quantum Basics

1.1 Second Quantization

1.1.1 One-particle basis functions

Any basis of a space must span the space and consist of linearly independent elements of that space. The formulation here is restricted to finite-dimensional vector/function spaces equipped with an inner product. In practice, this will be the case anyway, and the formulation is simpler this way. We can formulate the condition of $\{|\varphi_n\rangle\}$ spanning the space Φ by requiring, for all $|\varphi\rangle \in \Phi$, that there exists a set of coefficients $\{c_1, \dots, c_N\}$ such that

$$|\varphi\rangle = \sum_{n=1}^N c_n |\varphi_n\rangle = (|\varphi_1\rangle \cdots |\varphi_N\rangle) \begin{pmatrix} c_1 \\ \vdots \\ c_N \end{pmatrix}. \quad (1.1)$$

We can find a linear system of equations for the coefficients by considering

$$\langle \varphi_m | \varphi \rangle = \sum_{n=1}^N c_n \langle \varphi_m | \varphi_n \rangle, \quad (1.2)$$

so the linear system of equations becomes

$$\begin{pmatrix} \langle \varphi_1 | \varphi \rangle \\ \vdots \\ \langle \varphi_N | \varphi \rangle \end{pmatrix} = \begin{pmatrix} \langle \varphi_1 | \varphi_1 \rangle & \cdots & \langle \varphi_1 | \varphi_N \rangle \\ \vdots & & \vdots \\ \langle \varphi_N | \varphi_1 \rangle & \cdots & \langle \varphi_N | \varphi_N \rangle \end{pmatrix} \begin{pmatrix} c_1 \\ \vdots \\ c_N \end{pmatrix}, \quad (1.3)$$

with the Gram matrix $\mathbf{S}$:

$$\mathbf{S} = \begin{pmatrix} \langle \varphi_1 | \varphi_1 \rangle & \cdots & \langle \varphi_1 | \varphi_N \rangle \\ \vdots & & \vdots \\ \langle \varphi_N | \varphi_1 \rangle & \cdots & \langle \varphi_N | \varphi_N \rangle \end{pmatrix}. \quad (1.4)$$

Solving for $\vec{c}$, we find

$$\vec{c} = \mathbf{S}^{-1} \begin{pmatrix} \langle \varphi_1 | \varphi \rangle \\ \vdots \\ \langle \varphi_N | \varphi \rangle \end{pmatrix}, \quad (1.5)$$

and inserting $\vec{c}$ back into (1.1), we find

$$|\varphi\rangle = \sum_n \sum_m |\varphi_n\rangle \left(\mathbf{S}^{-1}\right)_{nm} \langle \varphi_m | \varphi \rangle \quad (1.6)$$

and therefore

$$\sum_n \sum_m |\varphi_n\rangle \left(\mathbf{S}^{-1}\right)_{nm} \langle \varphi_m | = \mathbb{1}. \quad (1.7)$$

Proving the Gram matrix is invertible

$\mathbf{S}$ invertible implies, for a finite-dimensional space, that the kernel must be trivial:

$$\mathbf{S}\vec{a} = 0 \iff \vec{a} = 0. \quad (1.8)$$

Consider

$$\begin{pmatrix} \langle \varphi_1 | \varphi_1 \rangle & \cdots & \langle \varphi_1 | \varphi_N \rangle \\ \vdots & & \vdots \\ \langle \varphi_N | \varphi_1 \rangle & \cdots & \langle \varphi_N | \varphi_N \rangle \end{pmatrix} \begin{pmatrix} a_1 \\ \vdots \\ a_N \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}. \quad (1.9)$$

Then, for all $m = 1, \dots, N$,

$$\left\langle \varphi_m \left| \sum_{n=1}^N a_n \varphi_n \right. \right\rangle = 0 \quad (1.10)$$

holds. In words: there is a vector $|\varphi_a\rangle = \sum_{n=1}^N a_n |\varphi_n\rangle$ which lies in Φ and is yet orthogonal to each vector $|\varphi_m\rangle$, $m = 1, \dots, N$, which span the space and thereby also to itself. The only vector that fulfils this requirement is the zero vector. Therefore we find the new condition:

$$\sum_{n=1}^N a_n |\varphi_n\rangle \stackrel{!}{=} 0. \quad (1.11)$$

Since the vectors of the set $\{|\varphi_n\rangle\}_{n=1}^N$ are linearly independent, this equation is only satisfied by $\vec{a} = \vec{0}$. Meaning that the kernel is trivial, which implies $\mathbf{S}$ is invertible.

Going to infinite dimensions

In the infinite-dimensional case the requirements have to be formulated more carefully. The kernel being trivial will not suffice for the operator $\mathbf{S}$ to be invertible: since the operator is now a map between two countably infinite-dimensional spaces, the infinite-matrix picture for $\mathbf{S}$ remains valid, but $\mathbf{S}$ must additionally fulfil the property that its spectrum be bounded from above and below. In the finite-dimensional case, boundedness from below and trivial kernel are equivalent. An infinite-dimensional example of a non-invertible operator with trivial kernel is

$$\mathbf{S} = \begin{pmatrix} 1 & & & 0 \\ & \frac{1}{2} & & \\ & & \frac{1}{3} & \\ 0 & & & \ddots \end{pmatrix}. \quad (1.12)$$

The kernel is trivial, but there exists no $A > 0$ such that for all $c \in \ell^2$,

$$A\|c\| \leq \|\mathbf{S}c\|. \quad (1.13)$$

This means the range that $\mathbf{S}$ maps onto is *not closed*, and the inverse operator has diverging entries.

To see the problem with unbounded operators like $\mathbf{S}^{-1}$ in this example:

$$\mathbf{S}^{-1} = \begin{pmatrix} 1 & & & \\ & 2 & & \\ & & 3 & \\ & & & \ddots \end{pmatrix} \quad (1.14)$$

Consider the sequence $a_n = (1/n)_{n \in \mathbb{N}}$, which lies within ℓ^2 since $\sum_{n \in \mathbb{N}} (1/n)^2$ converges. However,

$$\mathbf{S}^{-1}a = (1, 1, 1, \dots), \quad (1.15)$$

where $\sum_{n \in \mathbb{N}} 1^2 \rightarrow \infty$ is not finite. The operator $\mathbf{S}^{-1}$ therefore maps a sequence that lies within ℓ^2 out of ℓ^2 .

The bra is not the dual element

Let $\{|\varphi_n\rangle\}$ be the set of basis vectors of a vector space V . Then the elements of the dual set $\{\langle\tilde{\varphi}_n|\}$ have the property:

$$\langle \tilde{\varphi}_m | \varphi_n \rangle = \delta_{mn}. \quad (1.16)$$

This property is obviously not fulfilled by the definition of the bra's (only for orthogonal vectors). We can find the dual elements via equation (??) to be:

$$\langle \tilde{\varphi}_m | = \sum_{m'} (\mathbf{S}^{-1})_{mm'} \langle \varphi_{m'} |. \quad (1.17)$$

The bra $\langle \varphi_m |$ is still an element of the dual space, however with:

$$\langle \varphi_m | : V \rightarrow \mathbb{K} \quad (1.18)$$

$$|\varphi\rangle \mapsto \int dx \varphi_m^*(x) \varphi(x). \quad (1.19)$$

For an orthogonal basis, this definition agrees with the definition of the dual element, for that reason it often goes unnoticed that $\langle \varphi_m |$ is not in general the dual element to $|\varphi_m\rangle$.

The position and momentum bra's and ket's

Consider the definition of the scalar product of two wave functions as demonstrated in (1.19). We can interpret $\varphi(x)$ as the result of a linear functional X_{LF} mapping φ to its value $\varphi(x)$ at x :

$$X_{LF} : \mathcal{S} \rightarrow \mathbb{C} \quad (1.20)$$

$$\varphi \mapsto \varphi(x) \quad (1.21)$$

$\mathcal{S}$ just denotes the physical subspace of the Hilbert space L^2 of square-integrable functions [1]. Similarly, we can interpret $\varphi_m^*(x)$ as the result of the anti-linear functional X_{AF} mapping φ_m to the complex conjugate of its value $\varphi_m^*(x)$ at x :

$$X_{AF} : \mathcal{S} \rightarrow \mathbb{C} \quad (1.22)$$

$$\varphi \mapsto \varphi^*(x) \quad (1.23)$$

With this interpretation we can rewrite the inner product:

$$\langle \varphi | \psi \rangle = \int dx \varphi^*(x) \psi(x) = \int dx X_{AF}(\varphi) X_{LF}(\psi). \quad (1.24)$$

Now we define the ket $|x\rangle$ to be the anti-linear functional and the bra $\langle x|$ to be the linear functional, since the inner product here is defined to be anti-linear in the first entry and linear in the second entry. Therefore we can rewrite (1.24) to:

$$\langle \varphi | \psi \rangle = \int dx \langle \varphi | x \rangle \langle x | \psi \rangle. \quad (1.25)$$

Further we find, if $\varphi(x)$ and $\psi(x)$ are connected via a Fourier transform with $\varphi(p)$ and $\psi(p)$ respectively, then

$$\int dx \varphi^*(x) \psi(x) = \int dp \varphi^*(p) \psi(p). \quad (1.26)$$

Proof.

$$\int dx \varphi^*(x) \psi(x) = \int dx \int dp \frac{1}{\sqrt{2\pi\hbar}} e^{-ipx/\hbar} \varphi^*(p) \int dp' e^{ip'x/\hbar} \frac{1}{\sqrt{2\pi\hbar}} \psi(p') \quad (1.27)$$

$$= \int dp dp' \varphi^*(p) \psi(p') \frac{1}{2\pi\hbar} \int dx e^{ix(p'-p)/\hbar} \quad (1.28)$$

$$= \int dp dp' \varphi^*(p) \psi(p') \delta(p - p') \quad (1.29)$$

$$= \int dp \varphi^*(p) \psi(p). \quad (1.30)$$

□

So with the same arguments used for the position bra and ket, we find for momentum:

$$\langle \varphi | \psi \rangle = \int dp \langle \varphi | p \rangle \langle p | \psi \rangle. \quad (1.31)$$

If we consider the Fourier relationship between $\varphi(x)$ and $\varphi(p)$, we can find an additional interpretation of the position and momentum bra's as plane waves:

$$\langle x | \varphi \rangle = \varphi(x) = \int dp \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar} \varphi(p). \quad (1.32)$$

At the same time we can interpret the scalar product over two momentum-dependent functions directly:

$$\langle x | \varphi \rangle = \int dp x^*(p) \varphi(p) = \int dp \langle x | p \rangle \langle p | \varphi \rangle, \quad (1.33)$$

so we find

$$\langle p|x\rangle = x(p) = \frac{1}{\sqrt{2\pi\hbar}} e^{-ipx/\hbar} \quad (1.34)$$

and analogously

$$p(x) = \langle x|p\rangle = \langle p|x\rangle^* = \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar}. \quad (1.35)$$

We can also use the plane-wave interpretation to determine the overlaps $\langle x|x'\rangle$ and $\langle p|p'\rangle$ by directly computing the inner product of the plane-wave functions:

$$\langle x|x'\rangle = \int dp \frac{1}{2\pi\hbar} e^{ipx/\hbar} e^{-ipx'/\hbar} = \delta(x - x') \quad (1.36)$$

$$\langle p|p'\rangle = \int dx \frac{1}{2\pi\hbar} e^{-ipx/\hbar} e^{ip'x/\hbar} = \delta(p - p'). \quad (1.37)$$

At this point I should state that there are also approaches [1] which determine various "inner products" of functionals purely as solutions to differential equations. Also a very nice approach, which remains loyal to the functional interpretation. The only downside is, that one is stuck with many bra-kets which are not understood as inner products.

Note: That being said the careful reader probably realized that the bra-ket $\langle x|p\rangle$ and its complex conjugate can also not be understood as an inner product. In this derivation we find its value via the fourier relations. Also the interpretation of the position and momentum bra's and ket's as functionals is a preliminary to finding the inner product interpretation later. One can also interpret the plane-waves as kernels to the functionals.

The completeness relation in position and momentum representation

In the following, α can represent position x or momentum p . We consider the completeness relation (1.6) in α -representation and expand the inner product in the α -representation:

$$\langle \alpha|\varphi\rangle = \sum_n \sum_m \langle \alpha|\varphi_n\rangle \left(\mathbf{S}^{-1}\right)_{nm} \int d\alpha' \langle \varphi_m|\alpha'\rangle \langle \alpha'|\varphi\rangle \quad (1.38)$$

$$\varphi(\alpha) = \sum_n \sum_m \varphi_n(\alpha) \left(\mathbf{S}^{-1}\right)_{nm} \int d\alpha' \varphi_m^*(\alpha') \varphi(\alpha') \quad (1.39)$$

So we find, within an integral over α' :

$$\sum_n \sum_m \varphi_n(\alpha) \left(\mathbf{S}^{-1}\right)_{nm} \varphi_m^*(\alpha') = \delta(\alpha - \alpha'). \quad (1.40)$$

We can also consider the inner product in momentum representation

$$\langle x|\varphi\rangle = \sum_n \sum_m \langle x|\varphi_n\rangle \left(\mathbf{S}^{-1}\right)_{nm} \int dp \langle \varphi_m|p\rangle \langle p|\varphi\rangle \quad (1.41)$$

$$= \int dp \sum_n \sum_m \varphi_n(x) \left(\mathbf{S}^{-1}\right)_{nm} \varphi_m^*(p) \varphi(p) \quad (1.42)$$

and find the mixed representation completeness relation

$$\sum_n \sum_m \varphi_n(x) \left(\mathbf{S}^{-1}\right)_{nm} \varphi_m^*(p) = \frac{1}{\sqrt{2\pi\hbar}} \exp(ipx/\hbar). \quad (1.43)$$

1.1.2 Orbital Creation and Annihilation operators

It is assumed that the diagonal entries of the Gram-Matrix are normalized. Ensuring normalization is practically not a problem and without normalized states the formulas become unnecessarily crowded.

Bosons:

$$[\hat{b}_i, \hat{b}_j] = 0, \quad [\hat{b}_i^\dagger, \hat{b}_j^\dagger] = 0 \quad (1.44)$$

Case 1: $[\hat{b}_i, \hat{b}_j^\dagger] = \delta_{ij}$

$$|n_1, \dots, n_L\rangle = \prod_{k=1}^L \frac{(\hat{b}_k^\dagger)^{n_k}}{\sqrt{n_k!}} |\text{vac}\rangle \quad (1.45)$$

Case 2: $[\hat{b}_i, \hat{b}_j^\dagger] = \mathbf{S}_{ij}$

$$|n_1, \dots, n_L\rangle = \frac{1}{\sqrt{\text{perm}(\mathbf{S}_{\text{occ}})}} \prod_{k=1}^L (\hat{b}_k^\dagger)^{n_k} |\text{vac}\rangle \quad (1.46)$$

The $\frac{1}{\sqrt{n_k!}}$ term is just a special case of the more general permanent formula (The permanent is contains the same terms as the determinant but all signs are plus signs). The factor ensures normalization of the state. This can easily be shown just using the commutation relations of the operators.

Fermions:

$$\{\hat{c}_i^\dagger, \hat{c}_j^\dagger\} = 0, \quad \{\hat{c}_i, \hat{c}_j\} = 0, \quad \{\hat{c}_i, \hat{c}_j^\dagger\} = \mathbf{S}_{ij}. \quad (1.47)$$

First-quantized/Orbital-list:

$$|\vec{k}\rangle = |k_1, \dots, k_N\rangle = \frac{1}{\sqrt{\det(\mathbf{S}_{\text{occ}})}} \prod_{i=1}^N \hat{c}_{k_i}^\dagger |\text{vac}\rangle. \quad (1.48)$$

More intuitive and useful for theoretical considerations.

Second-quantized/Occupation-number:

$$\hat{c}_i^\dagger |n_1, \dots, n_i, \dots, n_L\rangle = (-1)^{\sum_{j=1}^{i-1} n_j} (1 - n_i) |n_1, \dots, n_i + 1, \dots, n_L\rangle, \quad (1.49)$$

$$\hat{c}_i |n_1, \dots, n_i, \dots, n_L\rangle = (-1)^{\sum_{j=1}^{i-1} n_j} n_i |n_1, \dots, n_i - 1, \dots, n_L\rangle. \quad (1.50)$$

This notation is used in computational approaches. Here the normalization factor of $\frac{\det(\mathbf{S}_{\text{occ, after}})}{\det(\mathbf{S}_{\text{occ, before}})}$ is neglected.

The mixed anti-commutator for fermions

Postulating the existence of the anti-commutator immediately returns its value:

$$\begin{aligned} \langle \varphi_n | \varphi_{n'} \rangle &= \langle \text{vac} | \hat{c}_n \hat{c}_{n'}^\dagger | \text{vac} \rangle \\ &= -\langle \text{vac} | \hat{c}_{n'}^\dagger \hat{c}_n | \text{vac} \rangle + \{ \hat{c}_n, \hat{c}_{n'}^\dagger \} \end{aligned}$$

So we find

$$\{ \hat{c}_n, \hat{c}_{n'}^\dagger \} = \langle \varphi_n | \varphi_{n'} \rangle = (\mathbf{S})_{nn'} \quad (1.51)$$

We can also see that this definition returns the expected overlaps:

$$\begin{aligned} \langle \varphi_n \varphi_m | \varphi_{n'} \varphi_{m'} \rangle &= \langle \text{vac} | \hat{c}_n \hat{c}_m \hat{c}_{m'}^\dagger \hat{c}_{n'}^\dagger | \text{vac} \rangle \\ &= -\langle \text{vac} | \hat{c}_n \hat{c}_{m'}^\dagger \hat{c}_m \hat{c}_{n'}^\dagger | \text{vac} \rangle + \langle \text{vac} | \hat{c}_n \hat{c}_{n'}^\dagger | \text{vac} \rangle \langle \varphi_m | \varphi_{m'} \rangle \\ &= -\langle \text{vac} | \hat{c}_n \hat{c}_{m'}^\dagger | \text{vac} \rangle \langle \varphi_m | \varphi_{n'} \rangle + \langle \varphi_n | \varphi_{n'} \rangle \langle \varphi_m | \varphi_{m'} \rangle \\ &= -\langle \varphi_n | \varphi_{m'} \rangle \langle \varphi_m | \varphi_{n'} \rangle + \langle \varphi_n | \varphi_{n'} \rangle \langle \varphi_m | \varphi_{m'} \rangle \end{aligned}$$

For the Slater determinants we would find

$$\begin{aligned}
& \int dx dy \frac{1}{\sqrt{2}} \left(\varphi_n(x) \varphi_m(y) - \varphi_m(x) \varphi_n(y) \right)^* \frac{1}{\sqrt{2}} \left(\varphi_{n'}(x) \varphi_{m'}(y) - \varphi_{m'}(x) \varphi_{n'}(y) \right) \\
&= \frac{1}{2} \left[\langle \varphi_n | \varphi_{n'} \rangle \langle \varphi_m | \varphi_{m'} \rangle - \langle \varphi_n | \varphi_{m'} \rangle \langle \varphi_m | \varphi_{n'} \rangle - \langle \varphi_n | \varphi_{m'} \rangle \langle \varphi_m | \varphi_{n'} \rangle + \langle \varphi_n | \varphi_{n'} \rangle \langle \varphi_m | \varphi_{m'} \rangle \right] \\
&= \langle \varphi_n | \varphi_{n'} \rangle \langle \varphi_m | \varphi_{m'} \rangle - \langle \varphi_n | \varphi_{m'} \rangle \langle \varphi_m | \varphi_{n'} \rangle
\end{aligned}$$

If the sets $\{n, m\}$ and $\{n', m'\}$ contain the same elements/orbitals, say $n = n'$ and $m = m'$, we find:

$$\langle \varphi_n \varphi_m | \varphi_n \varphi_m \rangle = \langle \varphi_n | \varphi_n \rangle \langle \varphi_m | \varphi_m \rangle - \langle \varphi_n | \varphi_m \rangle \langle \varphi_m | \varphi_n \rangle \quad (1.52)$$

$$= \det(\mathbf{S}_{\text{occ}}) \quad (1.53)$$

where the subscript “occ” denotes that $\mathbf{S}_{\text{occ}}$ is the “small” Gram matrix of the occupied orbitals, not the one that contains the overlaps between all basis states:

$$\mathbf{S}_{\text{occ}} = \begin{pmatrix} \langle \varphi_n | \varphi_n \rangle & \langle \varphi_n | \varphi_m \rangle \\ \langle \varphi_m | \varphi_n \rangle & \langle \varphi_m | \varphi_m \rangle \end{pmatrix}. \quad (1.54)$$

If we had chosen $n = m'$ and $m = n'$, we would have found an additional minus sign, corresponding to $\text{sgn}(\sigma) = -1$ for the transposition.

To visualize the prefactor of each term, we first start talking about orbitals, because the annihilation operators correspond to bra-orbitals and the creation operators to ket-orbitals. Standard-ordering is achieved if we take the overlap of adjacent bra and ket orbitals always:

$$\begin{aligned}
\langle \varphi_n \varphi_m | \varphi_{n'} \varphi_{m'} \rangle &= \langle \text{vac} | \hat{c}_n \hat{c}_m \hat{c}_{m'}^\dagger \hat{c}_{n'}^\dagger | \text{vac} \rangle \\
&\Downarrow \\
&\left[(\langle \varphi_n | \langle \varphi_m |) (| \varphi_{m'} \rangle | \varphi_{n'} \rangle) \right] \\
&\quad \langle \varphi_m | \varphi_{m'} \rangle \left[\langle \varphi_n | \right] \left[| \varphi_{n'} \rangle \right] \\
&\quad \langle \varphi_n | \varphi_{n'} \rangle \langle \varphi_m | \varphi_{m'} \rangle
\end{aligned}$$

Whenever one anti-commutes with an operator/orbital first, before evaluating the anticommutator, that corresponds to an orbital appearing to the *left* of the orbital even though it should be to the right in standard-ordering. So the prefactor becomes $(-1)^{n_L}$, where n_L denotes the orbitals that are matched with an “earlier” (further to the left) bra-orbital than they would be in normal ordering — or equivalently, the number of ket-orbitals that should

be to the right of the corresponding ket-orbital. The product of all those factors represents the permutation-factor, as the number of permutations needed corresponds to the sum of the n_L .

Proof. That the number of permutations corresponds to the sum of n_L for each position.

Strategy: Start with the left-most position with $n_L > 0$ and commute the orbital through its left neighbors. The orbitals that n_L refers to must be direct neighbors, otherwise you didn't choose the left-most position/orbital. Now iterate the procedure until you have found standard-ordering.

Now we can also quickly convince ourselves that we find the same pattern for the overlap of two Slater determinants. First, for the bra Slater determinant we only consider the normal-ordered term:

$$\varphi_1(x_1) \cdots \varphi_N(x_N)$$

Therefore the sign can only come from the term of the Slater determinant and corresponds to $(-1)^n$, with n being the number of permutations required to reach normal-ordering. For the other bra terms we use the trick:

$$\langle \varphi_1 | \varphi'_1 \rangle \cdots \langle \varphi_j | \varphi'_i \rangle \cdots \langle \varphi_i | \varphi'_j \rangle \cdots \langle \varphi_N | \varphi'_N \rangle = \langle \varphi_1 | \varphi'_1 \rangle \cdots \langle \varphi_i | \varphi'_j \rangle \cdots \langle \varphi_j | \varphi'_i \rangle \cdots \langle \varphi_N | \varphi'_N \rangle$$

where $i < j$. In other words, we can move the permutations of the bra term onto the ket term, and we again find that the sign is just $(-1)^n$. So the behaviour is the same.

Note: Here, in the Slater-determinant case, one finds all possible combinations for each of the $N!$ bra terms; Also, notice that for the overlap of a Slater determinant with itself all possible combinations just become $\det(\mathbf{S}_{\text{occ}})$ where $\mathbf{S}_{\text{occ}}$ is built with all the occupied orbitals. however, through the normalization of bra- and ket-state we find the factor $\frac{1}{N! \det(\mathbf{S}_{\text{occ}})}$.

Field Operators

Fundamental property:

$$\hat{\Psi}^\dagger(\alpha)|\text{vac}\rangle = |\alpha\rangle \quad \text{and} \quad \langle \text{vac} | \hat{\Psi}(\alpha) = \langle \alpha |, \quad (1.55)$$

where $\alpha \in \{x, p\}$.

We can use (1.7), for which we demonstrated that it holds between any bra-ket $\langle \gamma | \dots | \gamma' \rangle$ with $\gamma, \gamma' \in \{x, p, \varphi_k\}$ (For the specific case of the eigenfunctions $\{\phi_k\}$ that (1.7) is an identity just acting on an element of the function space spanned by the eigenfunctions (1.6)). Therefore we can expand the anti-linear functional $|\alpha\rangle$ and evaluate against the $\{\varphi_k\}$, which span the space of test functions:

$$\langle \varphi_k | \hat{\Psi}^\dagger(\alpha) | \text{vac} \rangle = \langle \varphi_k | \sum_n \sum_m |\varphi_n \rangle \left(\mathbf{S}^{-1} \right)_{nm} \langle \varphi_m | \alpha \rangle \quad (1.56)$$

$$\langle \varphi_k | \hat{\Psi}^\dagger(\alpha) | \text{vac} \rangle = \langle \varphi_k | \sum_n \sum_m \hat{c}_n^\dagger | \text{vac} \rangle \left(\mathbf{S}^{-1} \right)_{nm} \varphi_m^*(\alpha) \quad (1.57)$$

Since both functionals agree on all basis vectors, they agree on the entire space of test functions. And since the operators on both sides consist only of creation operators, we find operator equivalence:

$$\boxed{\hat{\Psi}^\dagger(\alpha) = \sum_n \sum_m \left(\mathbf{S}^{-1} \right)_{nm} \varphi_m^*(\alpha) \hat{c}_n^\dagger} \quad (1.58)$$

Before taking the hermitian conjugate, note that $\mathbf{S}^{-1}$ is hermitian, since $\mathbf{S}$ is hermitian:

$$\mathbf{S} \mathbf{S}^{-1} = \mathbb{1} \implies \left(\mathbf{S}^{-1} \right)^\dagger \mathbf{S}^\dagger = \left(\mathbf{S}^{-1} \right)^\dagger \mathbf{S} = \mathbb{1} \implies \left(\mathbf{S}^{-1} \right)^\dagger = \mathbf{S}^{-1}, \quad (1.59)$$

so that $\left(\mathbf{S}^{-1} \right)_{nm}^* = \left(\mathbf{S}^{-1} \right)_{mn}$. Taking the hermitian conjugate of the operator equivalence and relabelling $n \leftrightarrow m$:

$$\boxed{\hat{\Psi}(\alpha) = \sum_n \sum_m \left(\mathbf{S}^{-1} \right)_{nm} \varphi_n(\alpha) \hat{c}_m} \quad (1.60)$$

Further, we found that $\int d\alpha |\alpha \rangle \langle \alpha|$ acts like an identity inside any bra-ket $\langle \gamma | \dots | \gamma' \rangle$ with $\gamma, \gamma' \in \{x, p, \varphi_k\}$. We use this to expand the eigenfunctions and again test against the eigenfunctions:

$$\langle \varphi_k | \hat{c}_n^\dagger | \text{vac} \rangle = \langle \varphi_k | \int d\alpha |\alpha \rangle \langle \alpha | \hat{c}_n^\dagger | \text{vac} \rangle \quad (1.61)$$

$$\langle \varphi_k | \hat{c}_n^\dagger | \text{vac} \rangle = \langle \varphi_k | \int d\alpha \hat{\Psi}^\dagger(\alpha) | \text{vac} \rangle \varphi_n(\alpha) \quad (1.62)$$

With the same arguments as before we find the operator identity:

$$\boxed{\hat{c}_n^\dagger = \int d\alpha \hat{\Psi}^\dagger(\alpha) \varphi_n(\alpha)} \quad (1.63)$$

and by taking the hermitian conjugate

$$\boxed{\hat{c}_n = \int d\alpha \varphi_n^*(\alpha) \hat{\Psi}(\alpha)} \quad (1.64)$$

Consistency check. Substituting the field expansion into the mode expansion returns the mode operator:

$$\int d\alpha \hat{\Psi}^\dagger(\alpha) \varphi_n(\alpha) = \sum_{k,m} (\mathbf{S}^{-1})_{km} \hat{c}_k^\dagger \underbrace{\int d\alpha \varphi_m^*(\alpha) \varphi_n(\alpha)}_{=S_{mn}} = \sum_k \delta_{kn} \hat{c}_k^\dagger = \hat{c}_n^\dagger. \quad (1.65)$$

With the relationship between the field operators and creation/annihilation operators, we can establish the anti-commutators:

$$\{\hat{\Psi}(\alpha), \hat{\Psi}^\dagger(\alpha')\} = \sum_n \sum_m \sum_{n'} \sum_{m'} (\mathbf{S}^{-1})_{nm} \varphi_n(\alpha) (\mathbf{S}^{-1})_{n'm'} \varphi_{m'}^*(\alpha') \{\hat{c}_m, \hat{c}_{n'}^\dagger\} \quad (1.66)$$

$$= \sum_n \sum_m \sum_{n'} \sum_{m'} (\mathbf{S}^{-1})_{nm} (\mathbf{S})_{mn'} (\mathbf{S}^{-1})_{n'm'} \varphi_n(\alpha) \varphi_{m'}^*(\alpha') \quad (1.67)$$

$$= \sum_n \sum_{m'} (\mathbf{S}^{-1})_{nm'} \varphi_n(\alpha) \varphi_{m'}^*(\alpha') \quad (1.68)$$

$$= \delta(\alpha - \alpha') \quad (1.69)$$

and for the mixed anti-commutators:

$$\{\hat{\Psi}(\alpha), \hat{c}_k^\dagger\} = \sum_n \sum_m (\mathbf{S}^{-1})_{nm} \varphi_n(\alpha) \{\hat{c}_m, \hat{c}_k^\dagger\} \quad (1.70)$$

$$= \sum_n \sum_m (\mathbf{S}^{-1})_{nm} (\mathbf{S})_{mk} \varphi_n(\alpha) \quad (1.71)$$

$$= \sum_n \delta_{nk} \varphi_n(\alpha) = \varphi_k(\alpha) \quad (1.72)$$

and

$$\{\hat{c}_k, \hat{\Psi}^\dagger(\alpha)\} = \sum_n \sum_m (\mathbf{S}^{-1})_{nm} \varphi_m^*(\alpha) \{\hat{c}_k, \hat{c}_n^\dagger\} \quad (1.73)$$

$$= \sum_m \delta_{km} \varphi_m^*(\alpha) = \varphi_k^*(\alpha) \quad (1.74)$$

We find the three anti-commutators:

$$\boxed{\{\hat{\Psi}(\alpha), \hat{\Psi}^\dagger(\alpha')\} = \delta(\alpha - \alpha')} \quad (1.75)$$

$$\boxed{\{\hat{\Psi}(\alpha), \hat{c}_k^\dagger\} = \varphi_k(\alpha)} \quad (1.76)$$

$$\boxed{\{\hat{c}_k, \hat{\Psi}^\dagger(\alpha)\} = \varphi_k^*(\alpha)} \quad (1.77)$$

The Slater Determinant in second quantization

This (anti-)commutation relation leads to a nice definition of the Slater determinant:

$$\Phi_{\alpha_1 \alpha_2 \dots \alpha_N}(x_1, x_2, \dots, x_N) = \frac{1}{\sqrt{N! \det(\mathbf{S}_{\text{occ}})}} \langle \text{vac} | \hat{\Psi}(x_N) \dots \hat{\Psi}(x_1) \hat{c}_{\alpha_N}^\dagger \dots \hat{c}_{\alpha_1}^\dagger | \text{vac} \rangle \quad (1.78)$$

The normalization factor is the same for the Slater determinant in second and first quantization and will therefore be neglected in the following proof, that the formula (1.78) correctly represents the Slater determinant in first quantization. The first nontrivial case we find for $N = 2$:

$$\hat{\Phi}_{12}(x_1, x_2) = \langle \text{vac} | \hat{\Psi}(x_1) \hat{\Psi}(x_2) \hat{c}_2^\dagger \hat{c}_1^\dagger | \text{vac} \rangle \quad (1.79)$$

$$= \left[\langle \text{vac} | \hat{\Psi}(x_1) \varphi_2(x_2) \hat{c}_1^\dagger | \text{vac} \rangle - \langle \text{vac} | \hat{\Psi}(x_1) \hat{c}_2^\dagger \hat{\Psi}(x_2) \hat{c}_1^\dagger | \text{vac} \rangle \right] \quad (1.80)$$

$$= \left[\langle \text{vac} | \varphi_1(x_1) \varphi_2(x_2) | \text{vac} \rangle - \langle \text{vac} | \varphi_2(x_2) \hat{c}_1^\dagger \hat{\Psi}(x_1) | \text{vac} \rangle \right. \\ \left. - \langle \text{vac} | \hat{\Psi}(x_1) \hat{c}_2^\dagger \varphi_1(x_2) | \text{vac} \rangle + \langle \text{vac} | \hat{\Psi}(x_1) \hat{c}_2^\dagger \hat{c}_1^\dagger \hat{\Psi}(x_2) | \text{vac} \rangle \right] \quad (1.81)$$

$$= \left[\varphi_1(x_1) \varphi_2(x_2) - \langle \text{vac} | \varphi_2(x_1) \varphi_1(x_2) | \text{vac} \rangle \right. \\ \left. + \langle \text{vac} | \varphi_1(x_2) \hat{c}_2^\dagger \hat{\Psi}(x_1) | \text{vac} \rangle \right] \quad (1.82)$$

$$= \left(\varphi_1(x_1) \varphi_2(x_2) - \varphi_1(x_2) \varphi_2(x_1) \right) \quad (1.83)$$

where the tilde $\tilde{\Phi}$ denotes that the normalization factor is neglected.

Induction step $N \rightarrow N + 1$:

$$\begin{aligned} \langle \text{vac} | \hat{\Psi}(x_1) \dots \hat{\Psi}(x_{N+1}) \hat{c}_{\alpha_{N+1}}^\dagger \dots \hat{c}_{\alpha_1}^\dagger | \text{vac} \rangle &= \varphi_{\alpha_{N+1}}(x_{N+1}) \langle \text{vac} | \hat{\Psi}(x_1) \dots \hat{\Psi}(x_N) \hat{c}_{\alpha_N}^\dagger \dots \hat{c}_{\alpha_1}^\dagger | \text{vac} \rangle \\ &\quad - \langle \text{vac} | \hat{\Psi}(x_1) \dots \hat{\Psi}(x_N) \hat{c}_{\alpha_{N+1}}^\dagger \hat{\Psi}(x_{N+1}) \hat{c}_{\alpha_N}^\dagger \dots \hat{c}_{\alpha_1}^\dagger | \text{vac} \rangle \\ &\quad \vdots \\ &= \sum_{i=1}^{N+1} \varphi_i(x_{N+1}) \cdot \text{SD}(\varphi_1, \dots, \varphi_{i-1}, \varphi_{i+1}, \dots, \varphi_{N+1}) (-1)^{N+1-i} \end{aligned} \quad (1.84)$$

This is just the recursive formula for a determinant if you, for example, imagine $\varphi_i(x_{N+1})$ for $i \in \{\alpha_1, \dots, \alpha_{N+1}\}$ to be the first row and $\varphi_i(x_N)$ for $i \in \{1, \dots, N+1\}$ the second, and so on. Then “developing” the determinant after the first row gives the formula above.

This result allows us to connect first and second quantization, or abstract representation and position representation, in the same way we connected it for the single-particle case, where $\varphi(x) = \langle x|\varphi\rangle$. Similarly, we find for the many-particle case the abstract general N -electron state:

$$\boxed{|\varphi\rangle = \sum_{\alpha_1 < \dots < \alpha_N} a_{\alpha_1, \dots, \alpha_N} \hat{c}_{\alpha_N}^\dagger \cdots \hat{c}_{\alpha_1}^\dagger |\text{vac}\rangle} \quad (1.85)$$

and applying the bra

$$\boxed{\langle x_1, \dots, x_N| = \langle \text{vac}| \hat{\Psi}(x_1) \cdots \hat{\Psi}(x_N)} \quad (1.86)$$

to the abstract N -electron state gives the position representation / first quantization: a linear combination of Slater determinants.

2 Green Function Theory

2.1 The mathematical Perspective:

For a linear differential Operator L with:

$$L_t x(t) = F(t)$$

where the subscript denotes that the operator acts on t .

One can take the inner product with a function $G(t, t')$ called a Green Function:

$$\begin{aligned}\langle G(t, t') | L_{t'} x(t') \rangle &= \langle G(t, t') | F(t') \rangle \\ \int_{-\infty}^{\infty} dt' G(t, t') [L_{t'} x(t')] &= \int_{-\infty}^{\infty} dt' G(t, t') F(t') \\ \int_{-\infty}^{\infty} dt' [L_{t'}^\dagger G(t, t')] x(t') &= \int_{-\infty}^{\infty} dt' G(t, t') F(t')\end{aligned}$$

assuming we can find a $G(t, t')$ such that

$$L_{t'}^\dagger G(t, t') = \delta(t - t'),$$

we have found the solution of the differential equation:

$$x(t) = \int_{-\infty}^{\infty} dt' G(t, t') F(t')$$

When taking the hermitian conjugate of L_t , we are making use of the linearity of L , since we have assumed a linear definition of the scalar product. And therefore:

$$\begin{aligned}\langle x | \hat{A}y + \hat{A}z \rangle &= \langle x | \hat{A}y \rangle + \langle x | \hat{A}z \rangle = \\ &= \langle \hat{A}^\dagger x | y \rangle + \langle \hat{A}^\dagger x | z \rangle \\ &= \langle \hat{A}^\dagger x | y + z \rangle = \langle x | \hat{A}(y + z) \rangle \\ \Rightarrow \hat{A} \text{ is linear if } \hat{A}^\dagger \text{ exists.}\end{aligned}$$

We can also insert the solution of $x(t)$ back into the differential equation:

$$\begin{aligned}
 L_t x(t) &= F(t) \\
 L_t \left[\int_{-\infty}^{\infty} dt' G(t, t') F(t') \right] &= F(t) \\
 \Rightarrow \text{Linearity} \quad \int_{-\infty}^{\infty} dt' L_t [G(t, t') F(t')] &= F(t) \\
 \int_{-\infty}^{\infty} dt' L_t G(t, t') F(t') &= F(t) \\
 \Rightarrow L_t G(t, t') &= \delta(t - t')
 \end{aligned}$$

and we find

$$\boxed{L_t = L_{t'}^\dagger} \quad (2.1)$$

2.2 Connection to Linear Response

Assume that the linear response function $\chi(t - t')$ is defined by the following relationship, where $F(t)$ is the input function and $x(t)$ is the response function:

$$\boxed{x(t) = \int_{-\infty}^{\infty} dt' \chi(t - t') F(t')} \quad (2.2)$$

Note: In this definition the theta function, which ensures that only previous input values influence the response, is included in the response function.

Assume further that there is a linear differential operator L_t such that $x(t)$ solves the differential equation:

$$L_t x(t) = F(t)$$

Then we find that the Green Function with

$$L_{t'}^\dagger G(t, t') = L_t G(t, t') = \delta(t - t')$$

is a linear response function in terms of 2.2:

$$\begin{aligned}
 L_t x(t) &= \int_{-\infty}^{\infty} L_t G(t, t') F(t') dt' \\
 L_t x(t) &= F(t) \quad \square
 \end{aligned}$$

So for linear differential operators we will consider response functions and Green Functions to be synonymous.

2.3 Linear Response Theory – Kubo Formula

Note: The following provides a brief introduction to linear response theory and the Kubo formula and closely follows the presentation in [2].

Consider a quantum system described by the (time independent) Hamiltonian H_0 in thermodynamic equilibrium. This leads to the following expectation value of an observable $\hat{A}$:

$$\langle \hat{A} \rangle = \frac{1}{Z_0} \sum_n \exp(-\beta E_n) \langle n | \hat{A} | n \rangle \quad \text{with } Z_0 = \sum_n \exp(-\beta E_n)$$

$$\text{and } \hat{H}_0 |n\rangle = E_n |n\rangle$$

Assume, starting at time t_0 the system behaves according to:

$$\hat{H} = \hat{H}_0 + \hat{V}(t)$$

We use the interaction picture to approximate the time evolution due to $\hat{V}(t)$ up to first order:

$$\hat{V}_{H_0}(t) = \exp(iH_0 t) \hat{V}(t) \exp(-iH_0 t)$$

$$\hat{n}(t) = \exp(iH_0 t) n(t)$$

then

$$\hat{n}(t) = \hat{U}_{H_0}(t, t_0) \hat{n}(t_0)$$

with

$$\hat{U}_{H_0}(t, t_0) = 1 - \frac{i}{\hbar} \int_{t_0}^t dt_1 \hat{V}_{H_0}(t_1) + \left(\frac{i}{\hbar} \right)^2 \int_{t_0}^t dt_1 \hat{V}_{H_0}(t_1) \int_{t_0}^{t_1} dt_2 \hat{V}_{H_0}(t_2) + \dots$$

which we approximate by

$$\hat{U}_{H_0}(t, t_0) \approx 1 - \frac{i}{\hbar} \int_{t_0}^t dt_1 \hat{V}_{H_0}(t_1)$$

for the regular $|n\rangle$, we find by the interaction picture relationship:

$$n(t) = \exp(-iH_0 t) \hat{U}_{H_0}(t, t_0) \exp(iH_0 t_0) n(t_0)$$

$$= \exp(-iH_0 t) \hat{U}_{H_0}(t, t_0) n(t_0) \exp(iE_n t_0)$$

with $n(t_0) = |n\rangle$.

And thus:

$$\begin{aligned}
 \langle \hat{A} \rangle(t) &= \frac{1}{Z_0} \sum_n \exp(-\beta E_n) \langle n(t) | \hat{A} | n(t) \rangle \\
 &= \frac{1}{Z_0} \sum_n \exp(-\beta E_n) \langle n | \hat{U}_{H_0}^\dagger(t, t_0) \exp(iH_0 t) \hat{A} \exp(-iH_0 t) \hat{U}_{H_0}(t, t_0) | n \rangle \\
 &= \frac{1}{Z_0} \sum_n \exp(-\beta E_n) \langle n | \left(1 + \frac{i}{\hbar} \int_{t_0}^t \hat{V}_{H_0}(t_1) dt_1 \right) \hat{A}_{H_0}(t) \\
 &\quad \left(1 - \frac{i}{\hbar} \int_{t_0}^t \hat{V}_{H_0}(t_1) dt_1 \right) | n \rangle \\
 &= \frac{1}{Z_0} \sum_n \exp(-\beta E_n) \langle n | \hat{A} | n \rangle - \frac{i}{\hbar} \int_0^t dt_1 \frac{1}{Z_0} \sum_n \exp(-\beta E_n) \\
 &\quad \langle n | \hat{A}_{H_0}(t) \hat{V}_{H_0}(t_1) - \hat{V}_{H_0}(t_1) \hat{A}_{H_0}(t) | n \rangle \\
 &= \langle \hat{A} \rangle - \frac{i}{\hbar} \int_0^t dt_1 \langle [\hat{A}_{H_0}(t), \hat{V}_{H_0}(t_1)] \rangle \\
 &= \langle \hat{A} \rangle - \frac{i}{\hbar} \int_0^\infty dt_1 \Theta(t - t_1) \langle [\hat{A}_{H_0}(t), \hat{V}_{H_0}(t_1)] \rangle
 \end{aligned}$$

where we find the retarded bosonic Green function:

$$\boxed{G_{A,V}^R(t, t_1) = -\frac{i}{\hbar} \Theta(t - t_1) \langle [\hat{A}_{H_0}(t), \hat{V}_{H_0}(t_1)] \rangle} \quad (2.3)$$

The Green function can be interpreted as a response function if one assumes:

$$\hat{V}(t) = F(t) \hat{V}'(t)$$

where $F(t) \in \mathbb{C}$ for all t . Then we find

$$-\frac{i}{\hbar} \int_0^\infty dt_1 \Theta(t - t_1) \langle [\hat{A}_{H_0}(t), \hat{V}'_{H_0}(t_1)] \rangle F(t_1) \quad (2.4)$$

for the integral. The result (2.4) can be interpreted as the retarded Green function being the response function to the input function $F(t)$.

2.4 Green Functions: Definitions

This defines the Green functions with the most commonly used operators as well as the common time translation invariance. The section is intentionally kept specific such that it can easily be used as a reference. The entire Green function section is loosely based on books on this matter. [2][3][4] And the very detailed phd thesis by Mahdi Pourfath [5].

Advanced Green Function

$$G_{\sigma\sigma'}^A(r, t; r', t') = (+i) \Theta(t' - t) \left\langle \left[\psi_\sigma(r, t), \psi_{\sigma'}^\dagger(r', t') \right]_\varepsilon \right\rangle \quad (2.5)$$

$$G_{i\sigma j\sigma'}^A(t; t') = (+i) \Theta(t' - t) \left\langle \left[\hat{C}_{i\sigma}(t), \hat{C}_{j\sigma'}^\dagger(t') \right]_\varepsilon \right\rangle \quad (2.6)$$

where $[\cdot, \cdot]_{-1} = [\cdot, \cdot]$ and $[\cdot, \cdot]_1 = \{\cdot, \cdot\}$.

Usually we have time translational invariance so we will be using:

$$G_{\sigma\sigma'}^A(r, r', t) = (+i) \Theta(-t) \left\langle \left[\psi_\sigma(r, t), \psi_{\sigma'}^\dagger(r') \right]_\varepsilon \right\rangle \quad (2.7)$$

$$G_{i\sigma j\sigma'}^A(t) = (+i) \Theta(-t) \left\langle \left[\hat{C}_{i\sigma}(t), \hat{C}_{j\sigma'}^\dagger \right]_\varepsilon \right\rangle \quad (2.8)$$

Retarded Green Function

$$G_{\sigma\sigma'}^R(r, t; r', t') = (-i) \Theta(t - t') \left\langle \left[\psi_\sigma(r, t), \psi_{\sigma'}^\dagger(r', t') \right]_\varepsilon \right\rangle \quad (2.9)$$

$$G_{i\sigma j\sigma'}^R(t; t') = (-i) \Theta(t - t') \left\langle \left[\hat{C}_{i\sigma}(t), \hat{C}_{j\sigma'}^\dagger(t') \right]_\varepsilon \right\rangle \quad (2.10)$$

$$G_{\sigma\sigma'}^R(r, r', t) = (-i) \Theta(t) \left\langle \left[\psi_\sigma(r, t), \psi_{\sigma'}^\dagger(r') \right]_\varepsilon \right\rangle \quad (2.11)$$

$$G_{i\sigma j\sigma'}^R(t) = (-i) \Theta(t) \left\langle \left[\hat{C}_{i\sigma}(t), \hat{C}_{j\sigma'}^\dagger \right]_\varepsilon \right\rangle \quad (2.12)$$

Greater Keldysh Green Function

$$G_{\sigma\sigma'}^>(r, t, r', t') = (-i) \left\langle \psi_\sigma(r, t) \psi_{\sigma'}^\dagger(r', t') \right\rangle \quad (2.13)$$

$$G_{i\sigma j\sigma'}^>(t, t') = (-i) \left\langle \hat{C}_{i\sigma}(t) \hat{C}_{j\sigma'}^\dagger(t') \right\rangle \quad (2.14)$$

$$G_{\sigma\sigma'}^>(r, r', t) = (-i) \left\langle \psi_\sigma(r, t) \psi_{\sigma'}^\dagger(r') \right\rangle \quad (2.15)$$

$$G_{i\sigma j\sigma'}^>(t) = (-i) \left\langle \hat{C}_{i\sigma}(t) \hat{C}_{j\sigma'}^\dagger \right\rangle \quad (2.16)$$

Lesser Keldysh Green Functions

$$G_{\sigma\sigma'}^<(r, t, r', t') = (+i) \left\langle \psi_{\sigma'}^\dagger(r', t') \psi_\sigma(r, t) \right\rangle \quad (2.17)$$

$$G_{i\sigma j\sigma'}^<(t, t') = (+i) \left\langle \hat{C}_{j\sigma'}^\dagger(t') \hat{C}_{i\sigma}(t) \right\rangle \quad (2.18)$$

$$G_{\sigma\sigma'}^<(r, r', t) = (+i) \left\langle \psi_{\sigma'}^\dagger(r') \psi_\sigma(r, t) \right\rangle \quad (2.19)$$

$$G_{i\sigma j\sigma'}^<(t) = (+i) \left\langle \hat{C}_{j\sigma'}^\dagger \hat{C}_{i\sigma}(t) \right\rangle \quad (2.20)$$

The expectation value in all Green function definitions is either taken with respect to the vacuum state $|0\rangle$, the ground state $|\psi_0\rangle$ or the thermal state $\sum_n \exp(-\beta E_n) |n\rangle$. The vacuum state expectation can only be used for specific cases, which will be explained in the section on the vacuum Green function.

Matsubara/Imaginary time Green Functions

$$G_{\sigma\sigma'}(r, \tau, r', \tau') = - \left\langle \hat{T} \psi_\sigma(r, \tau) \psi_{\sigma'}^\dagger(r', \tau') \right\rangle \quad (2.21)$$

$$G_{i\sigma j\sigma'}(\tau, \tau') = - \left\langle \hat{T} \hat{C}_{i\sigma}(\tau) \hat{C}_{j\sigma'}^\dagger(\tau') \right\rangle \quad (2.22)$$

$$\begin{aligned} G_{\sigma\sigma'}(r, r', \tau) &= - \left\langle \psi_\sigma(r, -i\tau) \psi_{\sigma'}^\dagger(r') \right\rangle \Theta(\tau) \\ &\quad + \varepsilon \left\langle \psi_{\sigma'}^\dagger(r') \psi_\sigma(r, -i\tau) \right\rangle \Theta(-\tau) \end{aligned} \quad (2.23)$$

$$\begin{aligned} G_{i\sigma j\sigma'}(\tau) &= - \left\langle \hat{C}_{i\sigma}(-i\tau) \hat{C}_{j\sigma'}^\dagger \right\rangle \Theta(\tau) \\ &\quad + \varepsilon \left\langle \hat{C}_{j\sigma'}^\dagger \hat{C}_{i\sigma}(-i\tau) \right\rangle \Theta(-\tau) \end{aligned} \quad (2.24)$$

where

$$\hat{A}(\tau) = \hat{S}^{-1}(\tau, 0) \hat{A} \hat{S}(\tau, 0)$$

$$\hat{S}(\tau, \tau_0) = \hat{T} \exp \left[- \int_{\tau_0}^{\tau} d\tau' \hat{A}(\tau') \right]$$

from

$$\frac{\partial}{\partial \tau} \left[\hat{S}^{-1}(\tau, \tau_0) \hat{S}(\tau, \tau_0) \right] = 0$$

we find

$$\frac{\partial}{\partial \tau} \hat{S}^{-1}(\tau, \tau_0) = \hat{S}^{-1}(\tau, \tau_0) \hat{A}(\tau)$$

combined with the boundary condition

$$\hat{S}^{-1}(\tau_0, \tau_0) = \mathbb{1}$$

we find

$$\hat{S}^{-1}(\tau, \tau_0) = \hat{T}^{(R)} \exp \left[+ \int_{\tau_0}^{\tau} d\tau' \hat{A}(\tau') \right]$$

Where the time ordering operator $\hat{T}^{(R)}$ orders the operators in the opposite way as $\hat{T}$. In the time translation invariant case the operators are the usual Heisenberg operators as before. Here the expectation value refers to the thermal one. Sometimes using the limit $T \rightarrow 0$ one also finds the ground state expectation value.

Note: For convenience, the prefactor $\frac{1}{\hbar}$ appearing in the Kubo formula is omitted. The Hamiltonian for fermions is even in the number of fermionic operators and thus acts like a bosonic operator under the time ordering operator.

2.5 The Vacuum Green Function

In the case of a non-interacting and time translationally invariant (fermionic) system, we can also take the expectation value with respect to the vacuum state. Then the retarded Green Function becomes:

$$\boxed{G_{i\sigma j\sigma'}^R(t) = (-i) \Theta(t) \langle \text{vac} | \hat{C}_{i\sigma} \exp(-i\hat{H}t) \hat{C}_{j\sigma'}^\dagger | \text{vac} \rangle} \quad (2.25)$$

The other term of the anticommutator disappears, since the destruction operator acts on the vacuum state. The phase factor disappears, since $\hat{H}|\text{vac}\rangle = 0|\text{vac}\rangle$ for a normal ordered Hamiltonian.

Let's argue that both the retarded ground state and vacuum Green Function are the same in the non-interacting case:

$$\begin{aligned} G_{i\sigma j\sigma'}^{R,gs}(t) = (-i) \Theta(t) & \left[\langle \psi_0 | \hat{C}_{i\sigma} \exp(-i\hat{H}t) \hat{C}_{j\sigma'}^\dagger | \psi_0 \rangle \exp(iE_0t) \right. \\ & \left. + \langle \psi_0 | \hat{C}_{j\sigma'}^\dagger \exp(i\hat{H}t) \hat{C}_{i\sigma} | \psi_0 \rangle \exp(-iE_0t) \right] \end{aligned} \quad (2.26)$$

For the sake of simplicity, assume $\hat{C}_j^\dagger$ creates a particle in the single particle eigenstate $|n_j\rangle$, where $\hat{H}|n_j\rangle = \varepsilon_j|n_j\rangle$. [If the creation operators would not be eigenstate creation operators, they could still be represented by linear combinations of the eigenstate creation operators and if the summands are equal the sums are equal.]

$$\begin{aligned} \langle \psi_0 | \hat{C}_{i\sigma} \exp(-i\hat{H}t) \hat{C}_{j\sigma'}^\dagger | \psi_0 \rangle \exp(iE_0t) &= \delta_{ij} \exp(-i(E_0 + \varepsilon_j)t) \exp(iE_0t) \delta_{\sigma\sigma'} \\ &= \delta_{ij} \exp(-i\varepsilon_j t) \delta_{\sigma\sigma'} \end{aligned}$$

where $\varepsilon_j > 0$ since the state was empty in the ground state.

$$\langle \psi_0 | \hat{C}_{j\sigma'}^\dagger \exp(i\hat{H}t) \hat{C}_{i\sigma} | \psi_0 \rangle \exp(-iE_0t) = \delta_{ij} \exp(-i\varepsilon_j t) \delta_{\sigma\sigma'}$$

where $\varepsilon_j < 0$ since the state was occupied in the ground state. And the chemical potential is set to zero. So overall we find:

$$G_{i\sigma j\sigma'}^{R,gs}(t) = (-i) \Theta(t) \delta_{ij} \delta_{\sigma\sigma'} \exp(-i\varepsilon_j t)$$

And for the vacuum Green function we find:

$$\begin{aligned} G_{i\sigma j\sigma'}^{R,vac} &= (-i) \Theta(t) \langle \text{vac} | \hat{C}_{i\sigma} \exp(-i\hat{H}t) \hat{C}_{j\sigma'}^\dagger | \text{vac} \rangle \\ &= (-i) \Theta(t) \delta_{ij} \delta_{\sigma\sigma'} \exp(-i\varepsilon_j t) \end{aligned}$$

Here the ε_j 's are the eigenenergies of all single-particle states. $\square$

Now we want to show that the retarded vacuum Green function 2.25, and therefore 2.26, is a Green function of the single-particle Schrödinger equation in the mathematical sense. Considering the Schrödinger equation in the right basis this is straightforward:

$$\begin{aligned} \forall n_j : \quad i \frac{\partial}{\partial t} \langle n_j | \psi(t) \rangle - \langle n_j | \hat{H} | \psi(t) \rangle &= 0 \\ \forall n_j : \quad i \frac{\partial}{\partial t} \psi_{n_j}(t) - \varepsilon_j \psi_{n_j}(t) &= 0 \end{aligned}$$

where the $|\psi(t)\rangle$ is a single-particle wave function and the $|n_j\rangle$ are the single-particle eigenstates of $\hat{H}$.

$$\left[i \frac{\partial}{\partial t} - \varepsilon_j \right] G_{i\sigma j\sigma'}^{R,vac}(t) = \delta(t) \delta_{ij} \delta_{\sigma\sigma'} \quad \square$$

2.6 A closer look at the temperature Green function: (Rickayzen[4])

The expectation value is taken with respect to the thermal state:

$$\begin{aligned}
 G_{AB}(\tau, \tau') &= -\Theta(\tau - \tau') \langle \hat{A}(-i\tau) \hat{B}(-i\tau') \rangle \\
 &\quad + \varepsilon \Theta(\tau' - \tau) \langle \hat{B}(-i\tau') \hat{A}(-i\tau) \rangle \\
 &= -\Theta(\tau - \tau') \text{tr} \left[\exp(-\beta H) \exp(\tau H) \hat{A} \exp(-\tau H) \right. \\
 &\quad \left. \exp(\tau' H) \hat{B} \exp(-\tau' H) \right] \frac{1}{Z} \\
 &\quad + \varepsilon \Theta(\tau' - \tau) \text{tr} \left[\exp(-\beta H) \exp(\tau' H) \hat{B} \exp(-\tau' H) \right. \\
 &\quad \left. \exp(\tau H) \hat{A} \exp(-\tau H) \right] \frac{1}{Z} \\
 &= -\Theta(\tau - \tau') \text{tr} \left[\exp(-\beta H) \exp(H(\tau - \tau')) \hat{A} \exp(-H(\tau - \tau')) \hat{B} \right] \frac{1}{Z} \\
 &\quad + \varepsilon \Theta(\tau' - \tau) \text{tr} \left[\exp(-\beta H) \hat{B} \exp(H(\tau - \tau')) \hat{A} \exp(-H(\tau - \tau')) \right] \frac{1}{Z} \\
 &= G_{AB}(\tau - \tau')
 \end{aligned}$$

For $0 < \tau \leq \beta$:

$$\begin{aligned}
 G_{AB}(\tau - \beta) &= \varepsilon \text{tr} \left[\exp(-\beta H) \hat{B} \exp(H(\tau - \beta)) \hat{A} \exp(-H(\tau - \beta)) \right] \frac{1}{Z} \\
 &= \varepsilon \text{tr} \left[\exp(-\beta H) \exp(\tau H) \hat{A} \exp(-\tau H) \hat{B} \right] \frac{1}{Z} \\
 &= -\varepsilon G_{AB}(\tau)
 \end{aligned}$$

So for fermions we find, that G is a 2β -periodic function. This derivation does not generalize to the non-equilibrium case (H dependent on (imaginary) time). Since a Hamiltonian that is dependent on imaginary time, but time independent of real time would not be analytic one can also say H is independent of time.

Since $G(\tau)$ is 2β periodic we know we can find a Fourier expansion and the smallest frequency must be $\omega_{\min} = \frac{\pi}{\beta}$. Further of course $G(\tau + \beta) = -G(\tau)$, which leads to

$$\boxed{\omega_k = \left(\frac{2k+1}{\beta} \right) \pi} \Rightarrow \exp(i\omega_k \beta) = \exp(i(2k+1)\pi) = -1 \quad \forall k.$$

Those frequencies are the so called Matsubara frequencies.

Note: One sometimes finds the combinations of imaginary time Green Function with the ground state expectation value and Matsubara frequencies where β represents a made up temperature. In the derivation we could already see that this separation is strictly speaking

incorrect. Only in the limit $\beta \rightarrow \infty$ the ground state and the Matsubara frequencies can be combined. In this limit the thermal expectation value becomes the ground state expectation value and the Matsubara frequencies become continuous. If one assumes a β for the Matsubara frequencies one also assumes a β for the expectation value. That being said, for very large β the approximation should be good enough.

2.7 Lehmann Representation

The Lehmann Representation can be used to verify calculations for small systems and offers an easy but laborious way to analyse Green functions. It reduces to a single insertion of an identity in the form of eigenfunctions of the Hamiltonian:

$$\begin{aligned}\hat{H}|\psi_n\rangle &= E_n|\psi_n\rangle \\ I &= \sum_n |\psi_n\rangle\langle\psi_n|\end{aligned}$$

where the $|\psi_n\rangle$ are all the eigenfunctions of the Hamiltonian and not just the ones for a specified occupation.

Application: Retarded Green Function

$$\begin{aligned}G_{A,B}^R(t) &= (-i)\Theta(t) \langle [\hat{A}(t), \hat{B}]_\varepsilon \rangle \\ &= (-i)\Theta(t) \left[\langle \exp(i\hat{H}t) \hat{A} \exp(-i\hat{H}t) \hat{B} \rangle \right. \\ &\quad \left. + \varepsilon \langle \hat{B} \exp(i\hat{H}t) \hat{A} \exp(-i\hat{H}t) \rangle \right] \\ &= (-i)\Theta(t) \left[\sum_n \langle \exp(i\hat{H}t) \hat{A} |\psi_n\rangle\langle\psi_n| \exp(-i\hat{H}t) \hat{B} \rangle \right. \\ &\quad \left. + \varepsilon \sum_n \langle \hat{B} \exp(i\hat{H}t) |\psi_n\rangle\langle\psi_n| \hat{A} \exp(-i\hat{H}t) \rangle \right] \\ &= (-i)\Theta(t) \left[\sum_n \exp(-iE_n t) \langle \exp(i\hat{H}t) \hat{A} |\psi_n\rangle\langle\psi_n| \hat{B} \rangle \right. \\ &\quad \left. + \varepsilon \sum_n \exp(iE_n t) \langle \hat{B} |\psi_n\rangle\langle\psi_n| \hat{A} \exp(-i\hat{H}t) \rangle \right]\end{aligned}$$

The expectation value is intentionally left unspecified. One can already see that the two exponentials of the Hamiltonian will also become numbers for all the choices of expectation values discussed so far. The thermal expectation value, more accurately, introduces an additional sum over the eigenfunctions.

Application: Imaginary Time Green Function

$$\begin{aligned}
 G_{AB}(\tau) &= -\langle \exp(\hat{H}\tau) \hat{A} \exp(-\hat{H}\tau) \hat{B} \rangle \Theta(\tau) \\
 &\quad + \varepsilon \langle \hat{B} \exp(\hat{H}\tau) \hat{A} \exp(-\hat{H}\tau) \rangle \Theta(-\tau) \\
 &= -\sum_n \exp(-E_n \tau) \Theta(\tau) \langle \exp(\hat{H}\tau) \hat{A} |\psi_n\rangle \langle \psi_n| \hat{B} \rangle \\
 &\quad + \varepsilon \sum_n \exp(E_n \tau) \Theta(-\tau) \langle \hat{B} |\psi_n\rangle \langle \psi_n| \hat{A} \exp(-\hat{H}\tau) \rangle
 \end{aligned}$$

Substituting the corresponding expectation values we find:

Ground state Retarded Green Function:

$$\begin{aligned}
 G_{AB}^R(t) &= (-i) \Theta(t) \left[\sum_n \exp(it(E_0 - E_n)) \langle \psi_0 | \hat{A} | \psi_n \rangle \langle \psi_n | \hat{B} | \psi_0 \rangle \right. \\
 &\quad \left. + \varepsilon \sum_m \exp(it(E_m - E_0)) \langle \psi_0 | \hat{B} | \psi_m \rangle \langle \psi_m | \hat{A} | \psi_0 \rangle \right]
 \end{aligned}$$

Thermal Retarded Green Function:

$$\begin{aligned}
 Z G_{AB}^R(t) &= (-i) \Theta(t) \left[\sum_n \sum_{n'} \exp(it(E_{n'} - E_n)) \exp(-\beta E_{n'}) \langle \psi_{n'} | \hat{A} | \psi_n \rangle \langle \psi_n | \hat{B} | \psi_{n'} \rangle \right. \\
 &\quad \left. + \varepsilon \sum_m \sum_{m'} \exp(it(E_m - E_{m'})) \exp(-\beta E_{m'}) \langle \psi_{m'} | \hat{B} | \psi_m \rangle \langle \psi_m | \hat{A} | \psi_{m'} \rangle \right] \\
 &= (-i) \Theta(t) \left[\sum_n \sum_{n'} \exp(it(E_{n'} - E_n)) \langle \psi_{n'} | \hat{A} | \psi_n \rangle \langle \psi_n | \hat{B} | \psi_{n'} \rangle (\exp(-\beta E_{n'}) + \varepsilon \exp(-\beta E_n)) \right]
 \end{aligned}$$

Ground state Imaginary Time Green Function:

$$\begin{aligned}
 G_{AB}(\tau) &= -\Theta(\tau) \sum_n \exp(\tau(E_0 - E_n)) \langle \psi_0 | \hat{A} | \psi_n \rangle \langle \psi_n | \hat{B} | \psi_0 \rangle \\
 &\quad + \varepsilon \Theta(-\tau) \sum_m \exp(\tau(E_m - E_0)) \langle \psi_0 | \hat{B} | \psi_m \rangle \langle \psi_m | \hat{A} | \psi_0 \rangle
 \end{aligned}$$

Thermal Imaginary Time Green Function: $(-\beta \leq \tau \leq \beta)$

$$\begin{aligned}
 Z G_{AB}(\tau) &= \sum_n \sum_{n'} \exp(\tau(E_{n'} - E_n)) \langle \psi_{n'} | \hat{A} | \psi_n \rangle \langle \psi_n | \hat{B} | \psi_{n'} \rangle \\
 &\quad \left[-\Theta(\tau) \exp(-\beta E_{n'}) + \varepsilon \Theta(-\tau) \exp(-\beta E_n) \right]
 \end{aligned} \tag{2.27}$$

Where Z is the partition function. In equation (2.27) the periodicity: $\lim_{\varkappa \rightarrow 0} G(0 - \varkappa) = -\lim_{\varkappa \rightarrow 0} G(\beta - \varkappa)$ becomes obvious.

One can also see the "periodicity", more accurately decay, for the ground state imaginary time Green Function, since E_0 is the energy of the ground state $(E_0 - E_n)$ will always be negative so for $\tau \rightarrow \infty$, $G_{AB} \rightarrow 0$ and $(E_m - E_0)$ will always be positive so for $\tau \rightarrow -\infty$, $G_{AB} \rightarrow 0$.

Most of the interesting properties of Green Functions require the Green Functions in the angular-frequency domain. For the imaginary time Green Functions, we have already discovered, that they are periodic, thus a Fourier Transform is already justified. For the retarded Green Function we instead use the Laplace transform to avoid the singularities on the real axis.

Retarded Green Function in Frequency Domain

→ Ground state:

$$\begin{aligned}
 G_{AB}^R(\omega) &= \int_0^\infty dt \exp((i\omega - \eta)t) (-i) \Theta(t) \left[\sum_n \exp(it(E_0 - E_n)) \langle \psi_0 | \hat{A} | \psi_n \rangle \langle \psi_n | \hat{B} | \psi_0 \rangle \right. \\
 &\quad \left. + \varepsilon \sum_m \exp(it(E_m - E_0)) \langle \psi_0 | \hat{B} | \psi_m \rangle \langle \psi_m | \hat{A} | \psi_0 \rangle \right] \\
 &= \sum_n \frac{\langle \psi_0 | \hat{A} | \psi_n \rangle \langle \psi_n | \hat{B} | \psi_0 \rangle}{\omega + i\eta + (E_0 - E_n)} + \varepsilon \sum_m \frac{\langle \psi_0 | \hat{B} | \psi_m \rangle \langle \psi_m | \hat{A} | \psi_0 \rangle}{\omega + i\eta + (E_m - E_0)} \\
 \boxed{G_{AB}^R(\omega) &= \sum_n \frac{\langle \psi_0 | \hat{A} | \psi_n \rangle \langle \psi_n | \hat{B} | \psi_0 \rangle}{\omega + i\eta + E_0 - E_n} + \varepsilon \sum_m \frac{\langle \psi_0 | \hat{B} | \psi_m \rangle \langle \psi_m | \hat{A} | \psi_0 \rangle}{\omega + i\eta + E_m - E_0}} \quad (2.28)
 \end{aligned}$$

If one considers the typical case where $\hat{A}$ is a destruction operator and $\hat{B}$ a creation operator, (2.28) has the property that particle excitations (first term) have singularities for positive real parts of ω and hole excitations (second term) have singularities only for negative real parts of ω . This property is lost when going to thermal expectation value.

→ Thermal:

$$\boxed{G_{AB}^R(\omega) = \frac{1}{Z} \sum_{n,m} \frac{\langle \psi_m | \hat{A} | \psi_n \rangle \langle \psi_n | \hat{B} | \psi_m \rangle}{\omega + i\eta + E_m - E_n} (\exp(-\beta E_m) + \varepsilon \exp(-\beta E_n))} \quad (2.29)$$

Imaginary Time Green Function in Frequency Domain:

Since the period differs for fermions and bosons, only the fermionic case will be considered $\varepsilon = +1$ with a period of 2β . The normalization constant $\frac{1}{\beta}$ is shifted to the inversetransform and the factor of 2, we get from integrating over half the period, is absorbed into the normalization constant.

→ Thermal:

$$\begin{aligned}
 G_{AB}(\omega_n) &= \int_0^\beta d\tau \sum_{n,m} \left[-\Theta(\tau) \exp(-\beta E_m) + \Theta(-\tau) \exp(-\beta E_n) \right] \\
 &\quad \exp(\tau(E_m - E_n)) \langle \psi_m | \hat{A} | \psi_n \rangle \langle \psi_n | \hat{B} | \psi_m \rangle \exp(i\omega_n \tau) \\
 &= \sum_{n,m} (-1) \int_0^\beta d\tau \exp(-\beta E_m) \exp(\tau(E_m - E_n)) C_{nm} \exp(i\omega_n \tau) \\
 &= \sum_{n,m} C_{nm} (-1) \exp(-\beta E_m) \left[\frac{1}{i\omega_n + E_m - E_n} \exp(\tau(E_m - E_n)) \exp(i\omega_n \tau) \right]_0^\beta \\
 &= \sum_{n,m} C_{nm} \frac{-\exp(-\beta E_m)}{i\omega_n + E_m - E_n} \left[\exp(\beta(E_m - E_n)) - 1 \right] \\
 &= \sum_{n,m} \frac{\langle \psi_m | \hat{A} | \psi_n \rangle \langle \psi_n | \hat{B} | \psi_m \rangle}{i\omega_n + E_m - E_n} \left[\exp(-\beta E_n) + \exp(-\beta E_m) \right] \\
 \boxed{G_{AB}(\omega_n) = \sum_{n,m} \frac{\langle \psi_m | \hat{A} | \psi_n \rangle \langle \psi_n | \hat{B} | \psi_m \rangle}{i\omega_n + E_m - E_n} \left[\exp(-\beta E_n) + \exp(-\beta E_m) \right]} &\quad (2.30)
 \end{aligned}$$

with

$$G_{AB}(\tau) = \frac{1}{\beta} \sum_n \exp(-i\omega_n \tau) G_{AB}(\omega_n) \quad n \in \mathbb{N}_0$$

→ Ground state:

$$\boxed{G_{AB}(\omega_n) = \sum_n \frac{\langle \psi_0 | \hat{A} | \psi_n \rangle \langle \psi_n | \hat{B} | \psi_0 \rangle}{i\omega - E_n + E_0} + \sum_m \frac{\langle \psi_0 | \hat{B} | \psi_m \rangle \langle \psi_m | \hat{A} | \psi_0 \rangle}{i\omega + E_m - E_0}} \quad (2.31)$$

where the Matsubara frequencies are now continuous.

Comparing (2.30) with (2.29) and (2.28) with (2.31) one can see, that in the frequency domain, the retarded and imaginary time Green Function can be obtained from one another by just replacing $\omega + i\eta$ by $i\omega_n$ (retarded → imaginary) or $i\omega_n$ by $\omega + i\eta$ (imaginary → retarded). Another way to view this, is that there is a general Green Function $G : \mathbb{C} \rightarrow \mathbb{C}$ and the retarded Green Function values are the values on the slightly shifted real axis while the Matsubara Green function values are the values on the imaginary axis.

So analytic continuation can be used (since both Green Functions are analytic) to obtain one from the other. In case of using the Lehmann representation to calculate the Green function: analytic continuation reduces to exchanging the variables, as mentioned before. Usually, for larger systems, one would not compute the Lehmann representation, at least not exactly, since that would require full diagonalization in the thermal case and diagonalization within the subspace of the ground state and the space spanned by states that overlap with $\hat{A}|\psi_0\rangle$, $\hat{A}^\dagger|\psi_0\rangle$, $\hat{B}|\psi_0\rangle$, $\hat{B}^\dagger|\psi_0\rangle$, in the ground state case.

2.8 Basis change for Green functions

The formula

$$G_{ij} = \sum_{\alpha\beta} a_{i\alpha} a_{j\beta}^* G_{\alpha\beta} \quad (2.32)$$

relates Green functions in different bases, where the operators are related like:

$$\hat{C}_i = \sum_{\alpha} a_{i\alpha} \hat{d}_{\alpha}$$

with $a_{i\alpha} = \langle \varphi_i | \psi_{\alpha} \rangle$ and

$$\begin{aligned} \hat{C}_i^+ |\text{vac}\rangle &= |\varphi_i\rangle \\ \hat{d}_{\alpha}^+ |\text{vac}\rangle &= |\psi_{\alpha}\rangle \end{aligned}$$

To verify the formula one just has to insert it into the definitions of the Green functions and since scalars commute with everything one immediately finds the identity in all cases.

2.9 Spectral Function

Consider a function of the following form:

$$f(\omega + i\eta) = \frac{1}{\omega + i\eta - \omega'}$$

then

$$\text{Im } f(\omega + i\eta) = -\frac{\eta}{(\omega - \omega')^2 + \eta^2}$$

And $\lim_{\eta \rightarrow 0} \text{Im } f(\omega + i\eta)$ like a delta function goes to infinity for $\omega = \omega'$ and is zero otherwise. So if we enforce the normalization condition:

$$-\int_{-\infty}^{\infty} \lim_{\eta \rightarrow 0} \text{Im } f(\omega + i\eta) d\omega = 1$$

then $-\lim_{\eta \rightarrow 0} \text{Im } f(\omega + i\eta)$ is equivalent to the delta function.

$$\begin{aligned} -\int_{-\infty}^{\infty} \lim_{\eta \rightarrow 0} \text{Im } f(\omega + i\eta) d\omega &= \lim_{\eta \rightarrow 0} \int_{-\infty}^{\infty} \frac{\eta}{(\omega - \omega')^2 + \eta^2} d\omega = \\ &= \lim_{\eta \rightarrow 0} \lim_{\omega'' \rightarrow \infty} \left[\arctan\left(\frac{\omega''}{\eta}\right) - \arctan\left(\frac{-\omega''}{\eta}\right) \right] = \pi \end{aligned}$$

One can find $\frac{d}{dy} \arctan(y)$ over the relationship $\frac{d}{dx} \arctan(\tan(x)) = 1$ with $y = \tan(x)$.

With the normalization constant, we find:

$$\delta(\omega - \omega') = -\frac{1}{\pi} \lim_{\eta \rightarrow 0} \text{Im} \left[\frac{1}{\omega + i\eta - \omega'} \right]$$

Using this relation we find for the imaginary part of the retarded Green Function, here with respect to the thermal expectation value:

$$-\frac{1}{\pi} \lim_{\eta \rightarrow 0} \text{Im} \left[G_{AB}^R(\omega + i\eta) \right] = \frac{1}{Z} \sum_{n,m} \langle \psi_m | \hat{A} | \psi_n \rangle \langle \psi_n | \hat{B} | \psi_m \rangle (\exp(-\beta E_m) + \varepsilon \exp(-\beta E_n)) \delta(\omega + E_m - E_n)$$

In the fermionic case this function is called the spectral function and usually denoted $A(\omega)$ or $S(\omega)$. In the bosonic case the weight $1 - \exp(-\beta\omega)$ is negative for $\omega < 0$ and positive for $\omega > 0$. The expression can therefore not be interpreted as a spectral function.

The retarded Green Function can also be computed with the spectral function:

$$G_{AB}^R(\omega + i\eta) = \int_{-\infty}^{\infty} d\omega' \frac{S(\omega')}{\omega + i\eta - \omega'} \quad (2.33)$$

This relation also holds in the bosonic case, but is not a relation between Green Function and spectral function.

2.10 Equations of Motion

The equation of motion method can be used to calculate or approximate Green functions analytically and is used in the Wick theorem. For the derivation I will be assuming single particle operators with the properties $[\hat{C}_i, \hat{C}_j^\dagger]_\varepsilon = \delta_{ij}$ and $[\hat{C}_i, \hat{C}_j]_\varepsilon = 0$. The derivations are performed only for imaginary time Green functions, but wherever the relationship between real and imaginary angular frequency from the Lehmann chapter applies, the results generalize accordingly. For the general imaginary time single-particle Green function

$$G_{ij}(\tau, \tau') = - \left\langle \hat{T} \hat{C}_i(\tau) \hat{C}_j^\dagger(\tau') \right\rangle$$

we find the derivative

$$\begin{aligned} \frac{\partial}{\partial \tau} G_{ij}(\tau, \tau') &= -\delta(\tau - \tau') \left\langle \hat{C}_i(\tau) \hat{C}_j^\dagger(\tau') + \varepsilon \hat{C}_j^\dagger(\tau') \hat{C}_i(\tau) \right\rangle \\ &\quad - \left\langle \hat{T} [\hat{H}, \hat{C}_i](\tau) \hat{C}_j^\dagger(\tau') \right\rangle \end{aligned}$$

→ Fourier Transform:

For the Fourier transform, we have to further restrict the systems under consideration, since the periodicity of the imaginary time Green function holds only for equilibrium systems. Further, the cases of the ground state and thermal expectation value have to be considered separately.

First, consider the left side of the equation for the ground state case:

$$\int_{-\infty}^{\infty} d\tau \left(\frac{\partial}{\partial \tau} G_{ij}(\tau) \right) \exp(i\omega\tau) = [G_{ij}(\tau) \exp(i\omega\tau)]_{-\infty}^{\infty} - i\omega G_{ij}(\omega)$$

where the first term is zero, because of the decay property of the ground state imaginary time Green function, which holds for bosons and fermions. The normalization factor 2π was carried over to the inverse transform.

For the thermal expectation value, we find:

$$\int_0^{\beta} d\tau \exp(i\omega_n\tau) \frac{\partial}{\partial \tau} G_{ij}(\tau) = [G_{ij}(\tau) \exp(i\omega_n\tau)]_0^{\beta} - i\omega_n G_{ij}(\omega_n)$$

where the first term disappears, since

$$G_{ij}(\beta) \exp(i\omega_n\beta) = G_{ij}(0) \exp(i\omega_n0),$$

for bosons and fermions. The normalization factor $\frac{1}{\beta}$ was again carried over to the inverse-transform.

The first term on the right-hand side of the differential equation becomes:

$$- \langle \hat{C}_i \hat{C}_j^{\dagger} + \varepsilon \hat{C}_j^{\dagger} \hat{C}_i \rangle = -\delta_{ij}$$

after the Fourier transform.

To treat the last term let $\hat{H}$ be a normal ordered Hamiltonian that preserves the particle number and introduce the notation:

$$\begin{aligned} \hat{C}_i^{\dagger} &= \hat{C}_{i_1 \dots i_n}^{\dagger} = \hat{C}_{i_1}^{\dagger} \dots \hat{C}_{i_n}^{\dagger} \\ \hat{C}_{\vec{j}} &= \hat{C}_{j_1 \dots j_n} = \hat{C}_{j_1} \dots \hat{C}_{j_n} \end{aligned}$$

then the general Hamiltonian can be written as:

$$\begin{aligned} \hat{H} &= \sum_{n=1}^N \sum_{\substack{i_1 \dots i_n \\ j_1 \dots j_n}} \hat{C}_{i_1 \dots i_n}^{\dagger} H_{i_1 \dots i_n, j_1 \dots j_n} \hat{C}_{j_1 \dots j_n} \\ &= \sum_{n=1}^N \sum_{\vec{i}, \vec{j}} \hat{C}_{\vec{i}}^{\dagger} H_{\vec{i}, \vec{j}} \hat{C}_{\vec{j}} \end{aligned}$$

Now let us consider a single n -body term of that Hamiltonian:

$$T_{\vec{i}, \vec{j}} = \hat{C}_{\vec{i}}^{\dagger} H_{\vec{i}, \vec{j}} \hat{C}_{\vec{j}}$$

and find the commutator

$$[T_{\vec{i}, \vec{j}}, \hat{C}_i]$$

in the case, that $\forall k \in [1, n] : i_k \neq i$ the commutator is zero for fermions and bosons, since the number of operators in $T_{\vec{i}, \vec{j}}$ is even.

Now consider the case $i_k = i$ for some $k \in [1, n]$, for fermions:

$$T_{\vec{i}, \vec{j}} \hat{C}_i = (-1)^{k-1} (-1)^{2n-1} \hat{C}_{i_k}^\dagger \hat{C}_i \hat{C}_{[1, n] \setminus k}^\dagger \hat{C}_{[1, n]} H_{[1, n] \setminus k \rightarrow i, [1, n]}$$

with $[1, n] \setminus k \rightarrow i = i_1 \dots i_{k-1} i i_{k+1} \dots i_n$ depending on the context $i_k \Leftrightarrow j_k$.

$$-\hat{C}_i T_{\vec{i}, \vec{j}} = (-1) (-1)^{k-1} \hat{C}_i \hat{C}_{i_k}^\dagger \hat{C}_{[1, n] \setminus k}^\dagger \hat{C}_{[1, n]} H_{[1, n] \setminus k \rightarrow i, [1, n]}$$

so we find

$$[T_{\vec{i}, \vec{j}}, \hat{C}_i] = (-1)^k \hat{C}_{[1, n] \setminus k}^\dagger H_{[1, n] \setminus k \rightarrow i, [1, n]} \hat{C}_{[1, n]}$$

In the bosonic case the result is the same if $i_k = i$ for only one $k \in [1, n]$, without the sign factor in front since the operators commute.

When we substitute the commutator back into the remaining part of the differential equation, we get:

$$\begin{aligned} & [T_{\vec{i}, \vec{j}}, \hat{C}_i] (\tau) \hat{C}_j^\dagger(\tau') \Theta(\tau - \tau') \\ & - \hat{C}_j^\dagger(\tau') [T_{\vec{i}, \vec{j}}, \hat{C}_i] (\tau) \Theta(\tau' - \tau) = \\ & = \hat{T} \left[\hat{C}_{i_1}^\dagger(\tau + (2n-2)\eta) \dots \hat{C}_{i_{k-1}}^\dagger(\tau + (2n-k)\eta) \hat{C}_j^\dagger(\tau') \right. \\ & \quad \hat{C}_{i_{k+1}}^\dagger(\tau + (2n-k-1)\eta) \dots \hat{C}_{i_n}^\dagger(\tau + n\eta) \hat{C}_{j_1}(\tau + (n-1)\eta) \dots \\ & \quad \left. \hat{C}_{j_n}(\tau) \right] H_{[1, n] \setminus k \rightarrow i, [1, n]} \\ & := -G_{[1, n] \setminus k \rightarrow j, [1, n]}(\tau + (2n-2)\eta, \dots, \tau + (2n-k)\eta, \tau', \tau + (2n-k-1)\eta, \dots, \tau) \\ & \quad H_{[1, n] \setminus k \rightarrow i, [1, n]} \end{aligned}$$

In the case $\tau > \tau'$ the time ordering operator has to use $(2n-k)$ permutations to move $\hat{C}_j^\dagger(\tau')$ to the end $(-1)^{2n-k}(-1)^k = 1$. And in the case $\tau' < \tau$ the time ordering operator has to use $(k-1)$ permutations to move $\hat{C}_j^\dagger(\tau')$ to the beginning, $(-1)^{k-1}(-1)^k = -1$.

η is chosen to be infinitesimal and just ensures the correct ordering of the operators.

$$- \left\langle \hat{T} [\hat{H}, \hat{C}_i](\tau) \hat{C}_j^\dagger(\tau') \right\rangle = \sum_{n=1}^N \sum_{k=1}^n \sum_{\substack{[1, n] \setminus k \\ [1, n]}} G_{[1, n] \setminus k \rightarrow j, [1, n]}(\tau', \tau) H_{[1, n] \setminus k \rightarrow i, [1, n]}$$

where (τ', τ) is just a shorthand for the long, correct expression. This highlights the problem one encounters in the equations of motion approach when handling many-body Hamiltonians.

For Hamiltonians with n -body operators, where $n \geq 2$, the imaginary time derivatives of the Green functions depend on ever increasing n -body Green functions.

Let's explicitly denote the result for a one-body and time translation invariant Hamiltonian:

$$\begin{aligned} -\left\langle \hat{T} [\hat{H}, \hat{C}_i](\tau) \hat{C}_j^\dagger \right\rangle &= \sum_{j_1} H_{i,j_1} G_{j_1 j}(0, \tau) \\ &= -\sum_{j_1} H_{i,j_1} G_{j_1 j}(\tau, 0) \end{aligned}$$

which yields the Fourier transform of the equation of motion:

$$\boxed{-z G_{ij}(z) = -\delta_{ij} - \sum_{j_1} H_{i,j_1} G_{j_1 j}(z)} \quad (2.34)$$

with $z = i\omega$ or $z = i\omega_n$.

2.11 The non-interacting SIAM

$$\hat{H} = \varepsilon_0 \hat{C}_0^\dagger \hat{C}_0 + \sum_{i=1}^N \varepsilon_i \hat{C}_i^\dagger \hat{C}_i + t_i \hat{C}_0^\dagger \hat{C}_i + t_i^* \hat{C}_i^\dagger \hat{C}_0$$

This is a specific case of the non-interacting SIAM, one, that I have been mostly working with, where the hopping terms from bath site to bath site are zero. Further, because of the non-interacting nature, the spins decouple. The Hamiltonian could be for either spin. That this form does not present a simplification you find immediately, if you assume a general bath and then diagonalize the bath operators (but not the impurity operators). The zero-site, in this notation, is the impurity site. To find the local impurity Green function $G_{00}(z)$, we can use the result (2.34):

$$\begin{aligned} z G_{00}(z) &= 1 + \sum_{j_1=0}^N H_{0j_1} G_{j_1 0}(z) \\ z G_{00}(z) &= 1 + \varepsilon_0 G_{00}(z) + \sum_{j_1=1}^N t_{j_1} G_{j_1 0}(z) \end{aligned} \quad (2.35)$$

where $G_{j_1 0}(z)$, $j_1 \in [1, N]$ again fulfills (2.34):

$$\begin{aligned} z G_{j_1 0}(z) &= \sum_{j_2=0}^N H_{j_1 j_2} G_{j_2 0}(z) \\ z G_{j_1 0}(z) &= t_{j_1}^* G_{00}(z) + \varepsilon_{j_1} G_{j_1 0}(z) \\ G_{j_1 0}(z) &= \frac{t_{j_1}^*}{z - \varepsilon_{j_1}} G_{00}(z) \end{aligned}$$

inserting back into (2.35) we find:

$$\left(z - \varepsilon_0 - \sum_{j_1=1}^N \frac{|t_{j_1}|^2}{z - \varepsilon_{j_1}} \right) G_{00}(z) = 1$$

$$\boxed{G_{00}(z) = \left[z - \varepsilon_0 - \sum_{j_1=1}^N \frac{|t_{j_1}|^2}{z - \varepsilon_{j_1}} \right]^{-1} = [z - \varepsilon_0 - \Delta(z)]^{-1}} \quad (2.36)$$

with hybridization function:

$$\boxed{\Delta(z) = \sum_{j_1=1}^N \frac{|t_{j_1}|^2}{z - \varepsilon_{j_1}}} \quad (2.37)$$

Alternative: Partial Matrix Inverse

Since $\hat{H}$ is a non-interacting Hamiltonian it would also suffice to consider the vacuum Green Function:

$$\begin{aligned} G_{00}(\tau) &= -\Theta(\tau) \langle \text{vac} | \hat{C}_0(\tau) \hat{C}_0^\dagger | \text{vac} \rangle \\ &= -\Theta(\tau) \langle \text{vac} | \hat{C}_0 \exp(-\hat{H}\tau) \hat{C}_0^\dagger | \text{vac} \rangle \\ &= -\Theta(\tau) \sum_{n,n'} \langle \psi_0 | \psi_n \rangle \langle \psi_n | \exp(-\hat{H}\tau) | \psi_{n'} \rangle \langle \psi_{n'} | \psi_0 \rangle \\ &= -\Theta(\tau) \sum_n \exp(-E_n \tau) \langle \psi_0 | \psi_n \rangle \langle \psi_n | \psi_0 \rangle \end{aligned}$$

where $\hat{H}|\psi_n\rangle = E_n|\psi_n\rangle$, $|\psi_n\rangle$ are the one-particle eigenfunctions.

$$\begin{aligned} G_{00}(i\omega) &= - \int_0^\infty d\tau \exp(i\omega\tau) \sum_n \exp(-E_n\tau) \langle \psi_0 | \psi_n \rangle \langle \psi_n | \psi_0 \rangle \\ &= \sum_n \frac{1}{i\omega - E_n} \langle \psi_0 | \psi_n \rangle \langle \psi_n | \psi_0 \rangle \\ &= \sum_n \langle \psi_0 | [i\omega \mathbb{1} - \hat{H}]^{-1} | \psi_n \rangle \langle \psi_n | \psi_0 \rangle \\ &= \langle \text{vac} | \hat{C}_0 [i\omega \mathbb{1} - \hat{H}]^{-1} \hat{C}_0^\dagger | \text{vac} \rangle \end{aligned}$$

So if we interpret $[i\omega \mathbb{1} - \hat{H}]^{-1}$ as a matrix $\underline{M}$, this corresponds to the entry $(\underline{M})_{00}$. This is consistent with the result of the equation of motion section (2.34).

For two matrices

$$\underline{M} = \begin{pmatrix} \underline{M}_{11} & \underline{M}_{12} \\ \underline{M}_{21} & \underline{M}_{22} \end{pmatrix}, \quad \underline{N} = \begin{pmatrix} \underline{N}_{11} & \underline{N}_{12} \\ \underline{N}_{21} & \underline{N}_{22} \end{pmatrix}$$

with

$$\underline{M} \cdot \underline{N} = \mathbb{1}$$

we find by solving the relationship for the $\underline{M}_{ij}$

$$\underline{M}_{11} = \left[\underline{N}_{11} - \underline{N}_{12} \underline{N}_{22}^{-1} \underline{N}_{21} \right]^{-1}$$

or with $\underline{N} = z \mathbb{1} - \underline{H}$ where $\hat{H} = \vec{C}^\dagger \underline{H} \vec{C}$ we find:

$$\begin{aligned} \left([z \mathbb{1} - \underline{H}]^{-1} \right)_{00} &= \left[(z - \varepsilon_0) - (t_1 \dots t_N) \begin{pmatrix} \frac{1}{z - \varepsilon_1} & & \\ & \ddots & \\ & & \frac{1}{z - \varepsilon_N} \end{pmatrix} \begin{pmatrix} t_1^* \\ \vdots \\ t_N^* \end{pmatrix} \right]^{-1} \\ &= \left[z - \varepsilon_0 - \sum_{i=1}^N \frac{|t_i|^2}{z - \varepsilon_i} \right]^{-1} \end{aligned}$$

Note: The singularities of a non-interacting Green function are located at the eigenvalues of $\underline{H}$. Therefore, it would be interesting if one could efficiently compute the eigenvalues of a hermitian matrix $\underline{H}$, by first determining the Green functions (2.34) directly via (2.34) and then Fourier transform the Green function on the real axis (singularities are ideal for Fourier transform) to time domain, where one could then use the prony method to fit the frequencies/eigenvalues.

2.12 Local Green function of the non-interacting Hubbard Model in 1D

$$\hat{H} = -t \sum_l \left(\hat{C}_l^\dagger \hat{C}_{l+1} + \hat{C}_{l+1}^\dagger \hat{C}_l \right) - \mu \sum_l \hat{C}_l^\dagger \hat{C}_l$$

with $t, \mu \in \mathbb{R}$. The Fourier Transform

$$\hat{C}_l = \frac{1}{\sqrt{N}} \sum_k \hat{d}_k \exp(ikl) \quad (2.38)$$

$$\hat{C}_l^\dagger = \frac{1}{\sqrt{N}} \sum_k \hat{d}_k^\dagger \exp(-ikl) \quad (2.39)$$

leads to the diagonal Hamiltonian:

$$\hat{H} = \sum_k (-2t \cos(k) - \mu) \hat{d}_k^\dagger \hat{d}_k = \sum_k \varepsilon_k \hat{d}_k^\dagger \hat{d}_k$$

with bandwidth $W = 4t$. We choose the eigenstates in k -space to be the product states, this choice is worth being mentioned, since the (k) -state and $(-k)$ -state have the same energy. The groundstate is the state with all $-\frac{\pi}{2} < k < \frac{\pi}{2}$ states occupied. And consider the Lehmann

representation of the ground state retarded Green function (2.28):

$$\begin{aligned} G_{k_1 k_2}^R(\omega) &= \sum_n \frac{\langle \psi_0 | \hat{C}_{k_1} | \psi_n \rangle \langle \psi_n | \hat{C}_{k_2}^\dagger | \psi_0 \rangle}{\omega + i\eta + E_0 - E_n} \\ &\quad + \sum_m \frac{\langle \psi_0 | \hat{C}_{k_2}^\dagger | \psi_m \rangle \langle \psi_m | \hat{C}_{k_1} | \psi_0 \rangle}{\omega + i\eta + E_m - E_0} \\ &= \delta_{k_1 k_2} \frac{1}{\omega + i\eta - \varepsilon_{k_1}} \end{aligned}$$

using the Green function basis change relation (2.32) and the basis change (2.38):

$$\begin{aligned} G_{l_1 l_2}^R(\omega) &= \sum_{k_1 k_2} a_{l_1 k_1} a_{l_2 k_2}^* G_{k_1 k_2}^R(\omega) \\ &= \sum_{k_1} \frac{1}{N} \exp(ik_1 l_1) \exp(-ik_1 l_2) \frac{1}{\omega + i\eta - \varepsilon_{k_1}} \\ &= \frac{1}{N} \sum_k \frac{1}{\omega + i\eta - \varepsilon_k} \exp(ik(l_1 - l_2)) \end{aligned}$$

and for the local Green function:

$$G_{ll}^R(\omega) = \frac{1}{N} \sum_k \frac{1}{\omega + i\eta - \varepsilon_k}$$

and using (2.33), we find the spectral function:

$$S(\omega) = \frac{1}{N} \sum_k \delta(\omega - \varepsilon_k)$$

Based on the discrete spectral function, we can also define its continuous analogon:

$$\begin{aligned} S(\omega) &:= \frac{1}{2\pi} \int_{-\pi}^{\pi} dk \delta(\omega - \varepsilon(k)) = \frac{1}{2\pi} \int_{-\pi}^{\pi} dk \delta(\omega + 2t \cos(k) + \mu) \\ &= \frac{1}{\pi} \int_0^{\pi} dk \delta(\omega + 2t \cos(k) + \mu) \end{aligned}$$

Substitution: $\sigma = \omega + 2t \cos(k)$ $\sigma_0 = \omega + 2t$

$$\begin{aligned} \frac{\partial \sigma}{\partial k} &= -\sqrt{4t^2 - (\sigma - \omega)^2} & \sigma_{\pi} &= \omega - 2t \\ &= \frac{1}{\pi} \int_{\omega-2t}^{\omega+2t} d\sigma \frac{1}{\sqrt{4t^2 - (\sigma - \omega)^2}} \delta(\sigma + \mu) \\ &= \frac{1}{\pi} \frac{1}{\sqrt{4t^2 - (\mu + \omega)^2}} \end{aligned}$$

Since the delta-function can only give a distribution when $-\mu - 2t \leq \omega \leq -\mu + 2t$, the formula only applies to the frequencies within that range.

3 Methods

3.1 Generalized Prony Method

This section very closely follows the excellent phd thesis of Thomas Peter [6]. It is assumed that every eigenspace V_j of the linear operator $\hat{A}$ is one-dimensional. If the eigenspace is not one-dimensional, we choose a representative. Thus for each eigenvalue λ_j there exists a unique eigenvector v_j .

Furthermore, we assume that the function f is M -sparse. Therefore, M eigenfunctions suffice to represent the function exactly:

$$f = \sum_{j \in J} c_j v_j \quad |J| = M$$

where J is a proper subset of the set of all eigenfunctions of the linear operator $\hat{A}$.

Due to the linearity of the operator $\hat{A}$ it holds:

$$\hat{A}^k f = \sum_{j \in J} c_j \lambda_j^k v_j$$

Now we assume a linear functional F , such that $Fv_j \neq 0$ holds for all $j \in I$. Then we can reconstruct f with $2M$ values:

$$F(\hat{A}^k f), \quad k = 0, \dots, 2M - 1.$$

To find this reconstruction, we start with the general definition of a polynomial whose roots are given by λ_j :

$$P(z) := \prod_{j \in J} (z - \lambda_j)$$

Alternatively, this polynomial can be represented as a sum of trivial polynomials:

$$P(z) = \sum_{k=0}^M p_k z^k$$

where the p_k are unknown and $p_M = 1$ is chosen. For $m \in \mathbb{N}$ we find:

$$\begin{aligned} \sum_{k=0}^M p_k F(\hat{A}^{k+m} f) &= \sum_{k=0}^M p_k F\left(\sum_{j \in J} c_j \lambda_j^{k+m} v_j\right) = \\ &= \sum_{j \in J} c_j \lambda_j^m \underbrace{\left(\sum_{k=0}^M p_k \lambda_j^k\right)}_{P(\lambda_j)=0} F v_j = 0 \end{aligned}$$

which yields with $p_M = 1$ the following linear system of equations for p_k :

$$\sum_{k=0}^{M-1} p_k F(\hat{A}^{k+m} f) = -F(\hat{A}^{M+m} f)$$

The matrix $\underline{H} = [F(\hat{A}^{k+m} f)]_{k,m}$ is invertible.

Proof:

$$\begin{aligned} \underline{H} &= [F(\hat{A}^{k+m} f)]_{k,m} = \left[\sum_{j \in J} c_j \lambda_j^{k+m} F(v_j) \right]_{k,m} \\ &= \left[\sum_{j \in J} \lambda_j^k c_j F(v_j) \lambda_j^m \right]_{k,m} \\ &= \begin{pmatrix} \lambda_1^0 & \cdots & \lambda_M^0 \\ \lambda_1^1 & \cdots & \lambda_M^1 \\ \vdots & & \vdots \\ \lambda_1^{M-1} & \cdots & \lambda_M^{M-1} \end{pmatrix} \begin{pmatrix} c_1 F(v_1) & & 0 \\ & \ddots & \\ 0 & & c_M F(v_M) \end{pmatrix} \begin{pmatrix} \lambda_1^0 & \cdots & \lambda_1^{M-1} \\ \vdots & & \vdots \\ \lambda_M^0 & \cdots & \lambda_M^{M-1} \end{pmatrix} \end{aligned}$$

By assumption, none of the c_j or $F(v_j)$ vanish, so the diagonal matrix is invertible. To show that the Vandermonde matrix

$$\underline{V}_\lambda := \left(\lambda_j^k \right)_{k=0, j \in J}^{M-1}$$

is also invertible, we use that for a square matrix invertibility is equivalent to linear independence of the rows and columns; We show it for the rows:

$$\sum_{k=0}^{M-1} c_k \lambda_j^k = 0 \quad \forall j \in J \quad \text{with } |J| = M$$

A polynomial of degree $(M-1)$ has at most $(M-1)$ roots $\Rightarrow$ the system of equations cannot be satisfied $\Rightarrow$ all row vectors are linearly independent $\Rightarrow$ the matrix is invertible.

The p_k are thus uniquely determined. Now we can find the eigenvalues λ_j as roots of the polynomial:

$$P(z) = \sum_{k=0}^M p_k z^k \quad \text{with } p_M = 1$$

and, for example, using a power method, we can find the eigenfunctions v_j .

Alternatively, one can solve the system of equations $(\hat{A} - \lambda_j)v_j = 0$ directly or minimize $v_j^\dagger (\hat{A} - \lambda_j)^\dagger (\hat{A} - \lambda_j) v_j$.

Finally, we only need to solve

$$\left[F(\hat{A}^k f) \right]_k = \left[\lambda_j^k \right]_{k,j \in J} \left[F(v_j) \right]_{j \in J} (c_j)_{j \in J}$$

for the vector $[F(v_j)c_j]_{j \in J}$ with $k = 0, \dots, 2M - 1$. So an overdetermined system of equations. Further, if we know $[F(v_j)]_{j \in J}$ we can find the coefficients $[c_j]_{j \in J}$.

3.1.1 Derivation of the Prony Method from the generalized Prony Method

The Prony method is obtained as a special case of the generalized Prony method by choosing $\hat{A} = \hat{T}(a) = \hat{T}_a$ and $F(f) = \int dx_0 \delta(x_0) f(x_0)$. Here

$$\hat{T}(a) f(x_0) = f(x_0 + a).$$

With these definitions for the functional and the operator, we find the following formula for the coefficients of the characteristic polynomial:

$$\begin{aligned} \sum_{k=0}^M p_k F(\hat{A}^{k+m} f) &= 0 \\ \sum_{k=0}^M p_k \int dx_0 \delta(x_0) \hat{T}_a^{k+m} f(x_0) &= 0 \\ \sum_{k=0}^M p_k f((k+m)a) &= 0 \end{aligned}$$

As a next step, we determine the roots of the characteristic polynomial / the eigenvalues of $\hat{T}_a$. In the case of the Prony method, we represent these as

$$\lambda_j = \exp(aT_j)$$

and thus find the eigenfunctions

$$\hat{T}_a \psi(x) = \exp(aT_j) \psi(x) \quad \Rightarrow \quad \psi(x) = \exp(T_j x).$$

To find the coefficients of the eigenfunction in the representation of $f(x)$, we only need to solve

$$F(\hat{A}^k f) = \sum_{j \in J} c_j \lambda_j^k F(v_j)$$

where

$$F(v_j) = \int dx_0 \delta(x_0) \exp(T_j x_0) = 1$$

and

$$F(\hat{A}^k f) = F(\hat{T}_a^k f) = \int dx_0 \delta(x_0) \hat{T}_a^k f(x_0) = f((k+1)a)$$

$$f(k) = \sum_{j \in J} c_j \lambda_j^k \quad k = 0, \dots, 2M-1$$

The Prony method solves the uniquely determined system of equations with $k = 0, \dots, M-1$ ($1, \dots, M$) and not the overdetermined one, as is the case with the generalized Prony method.

3.1.2 The Esprit Method

The standard Prony Method can only utilize $2M-1$ function evaluations, where M has to be known. Therefore, we introduce the Esprit method, which can incorporate more data and allows us to formulate an educated guess about the sparsity M . The idea is to replace the calculations of the p_i 's and instead directly calculate the zeros λ_j of the polynomial:

$$\sum_k p_k z^k \quad \text{with} \quad \sum_k p_k \lambda_j^k = 0$$

As in the case of the Prony Method, we have:

$$\sum_k p_k F(\hat{A}^{k+m} f(x)) = 0$$

Let $(\underline{H})_{mk} = F(\hat{A}^{k+m} f(x)) = h(k+m)$, then

$$\underline{H} \vec{p} = 0$$

and with $p_{M'} = 1$

$$\begin{pmatrix} h(0) & \cdots & h(M'-1) \\ \vdots & & \vdots \\ h(M'-1) & \cdots & h(2M'-2) \end{pmatrix} \begin{pmatrix} p_0 \\ \vdots \\ p_{M'-1} \end{pmatrix} = \begin{pmatrix} -h(M') \\ \vdots \\ -h(2M'-1) \end{pmatrix}$$

Here we use M' , since M' is an upper bound of the actual unknown sparsity M , so that we can use more data (in the case, where we know M). Here we see that the companion matrix to the polynomial C_p connects the following two matrices:

$$\underbrace{\begin{pmatrix} h(0) & \cdots & h(M'-1) \\ \vdots & & \vdots \\ h(M'-1) & \cdots & h(2M'-2) \end{pmatrix}}_{:=\underline{H}(0)} \underbrace{\begin{pmatrix} 0 & \cdots & -p_0 \\ 1 & 0 & \cdots & -p_1 \\ & \ddots & \ddots & \vdots \\ 0 & & 1 & -p_{M'-1} \end{pmatrix}}_{\substack{\text{the minus does not} \\ \text{affect the zeros}}} = \underbrace{\begin{pmatrix} h(1) & \cdots & h(M') \\ \vdots & & \vdots \\ h(M') & \cdots & h(2M'-1) \end{pmatrix}}_{:=\underline{H}(1)}$$

and the eigenvalues and eigenvectors of the companion matrix are also the eigenvalues and eigenvectors of the generalized eigenvalue problem:

$$[-\lambda_j \underline{H}(0) + \underline{H}(1)] \vec{v}_j = 0 \quad \Rightarrow \quad [-\lambda_j \underline{H}(0) + \underline{H}(0) \underline{C}_P] \vec{v}_j = 0$$

since $\underline{C}_P \vec{v}_j = \lambda_j \vec{v}_j$.

Solving the full generalized eigenproblem will, in general, give too many eigenvalues. Therefore, it is necessary to isolate the dominating components of the large matrix. This can be achieved efficiently using the singular value decomposition:

Let $\underline{H}$ denote the matrix, which contains all columns of $\underline{H}(0)$ and $\underline{H}(1)$ respectively:

$$\underline{H} = \begin{pmatrix} h(0) & \cdots & h(M') \\ \vdots & & \vdots \\ h(M' - 1) & \cdots & h(2M' - 1) \end{pmatrix} = \underline{U} \underline{S} \underline{V}$$

and let again $\underline{V}(0)$ denote the first M' columns of $\underline{V}$ and $\underline{V}(1)$ the last M' columns of $\underline{V}$. Then we find:

$$\underline{H}(0) = \underline{U} \underline{S} \underline{V}(0) \quad \underline{H}(1) = \underline{U} \underline{S} \underline{V}(1)$$

Assuming the sparsity M is known, the M rows of $\underline{V}$ and the M columns of $\underline{U}$ corresponding to the largest singular values are retained. The magnitude of the discarded singular values provide a measure of quality of the fit and the noise level present in the data.

If M is unknown, we can obtain a nice guess at this point, an initial estimate can be obtained by examining the singular value spectrum, for example by identifying a significant drop in the singular values or by using cutoff values.

The generalized eigenproblem can be simplified using the SVD expressions:

$$-\lambda_j \underline{H}(0) + \underline{H}(1) = -\lambda_j \underline{U} \underline{S} \underline{V}(0) + \underline{U} \underline{S} \underline{V}(1)$$

or

$$= -\lambda_j \underline{U}' \underline{S}' \underline{V}'(0) + \underline{U}' \underline{S}' \underline{V}'(1)$$

where the primes denote the reduced matrices.

$$\begin{aligned} (-\lambda_j \underline{U}' \underline{S}' \underline{V}'(0) + \underline{U}' \underline{S}' \underline{V}'(1)) \vec{v}_j &= \vec{0} \\ (-\lambda_j \underline{V}'(0) + \underline{V}'(1)) \vec{v}_j &= \vec{0} \\ \underline{V}'(0)^{-1} \underline{V}'(1) \vec{v}_j &= \lambda_j \vec{v}_j \end{aligned}$$

or formulated differently, we want to solve the eigenproblem:

$$\underline{A} \vec{v}_j = \lambda_j \vec{v}_j$$

where $\underline{A}$ solves:

$$\underline{V}'(0) \underline{A} = \underline{V}'(1)$$

This will give us the eigenvalues and from these we proceed as usual.

3.2 DMRG

The tensor notation is the notation developed by Jan von Delft and demonstrated in his lecture [7]. The derivations also follow at least loosely the derivations in the lecture.

DMRG is a Tensor Network Method for determining the ground state of a Hamiltonian. The method minimizes the energy $\langle \psi | \hat{H} | \psi \rangle$ under the constraint that the state remains normalized $\langle \psi | \psi \rangle = 1$. A straightforward approach is using a Lagrange multiplier and minimizing the cost function:

$$\langle \psi | \hat{H} | \psi \rangle - \lambda (\langle \psi | \psi \rangle - 1) \quad (3.1)$$

We want to minimize the above expression with respect to the tensors $[T_i]_{k_{i-1}, \sigma_i, k_i}$ where

$$[\psi]_{\sigma_1 \dots \sigma_L} = \sum_{k_2} \dots \sum_{k_L} [T_1]_{k_1 \sigma_1 k_2} \dots [T_L]_{k_L \sigma_L k_{L+1}}$$

k_1 and k_{L+1} are dummy indices and

$$|\psi\rangle = \sum_{\sigma_1} \dots \sum_{\sigma_L} [\psi]_{\sigma_1 \dots \sigma_L} |\sigma_1\rangle \otimes \dots \otimes |\sigma_L\rangle$$

with Hamiltonian

$$\hat{H} = \sum_{\{\sigma\}, \{\sigma'\}} [H]_{\sigma_1 \dots \sigma_L \sigma'_1 \dots \sigma'_L} |\sigma_1 \dots \sigma_L\rangle \langle \sigma'_1 \dots \sigma'_L|.$$

where the coefficients are given by:

$$[H]_{\sigma_1 \dots \sigma_L \sigma'_1 \dots \sigma'_L} = \sum_{k_2} \dots \sum_{k_L} [h_1]_{k_1 \sigma_1 \sigma'_1 k_2} \dots [h_L]_{k_L \sigma_L \sigma'_L k_{L+1}}$$

Although the symbol σ is commonly associated with spin degrees of freedom, here it denotes a general local physical index. For example, σ_j may represent a site occupation number and is not necessarily binary. Its range is determined by the dimension of the local physical Hilbert space.

In tensor notation, equation (3.1) takes the form:

Taking the derivative by $\frac{\partial}{\partial A_i^\dagger}$ we find 3.2, which is just a standard eigenvalue problem in graphical notation for the effective Hamiltonian 3.3.

The derivative by λ just leads to the auxiliary condition that the eigenvector of the eigenvalue problem has to be normalized. The updated tensor T_i is chosen as the (normalized) ground state of the effective Hamiltonian. A shift of the active site is performed by applying a QR-decomposition followed by a tensor contraction. The only heuristic choices in this algorithm are the ranks of the original state, which can not be increased and the choice of the ground

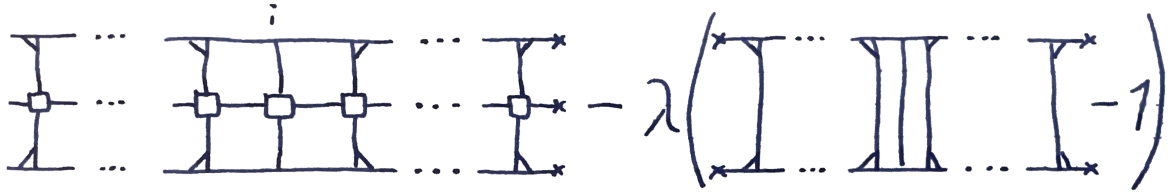


Figure 3.1: DMRG cost function $\langle \psi | \hat{H} | \psi \rangle - \lambda (\langle \psi | \psi \rangle - 1)$ in tensor-network form. The MPS is in site-canonical form with respect to the site i .

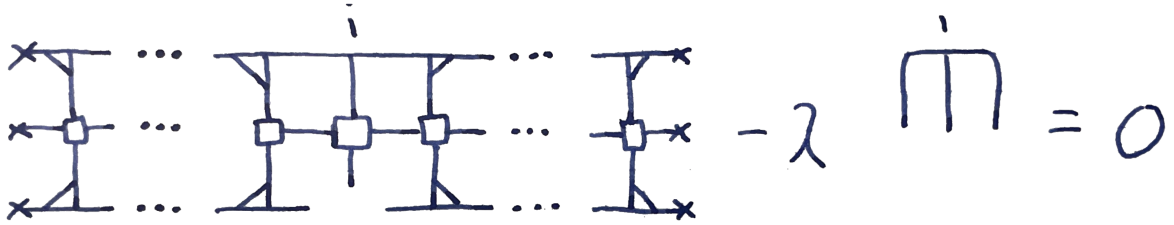


Figure 3.2: DMRG eigenvalue problem in graphical notation.

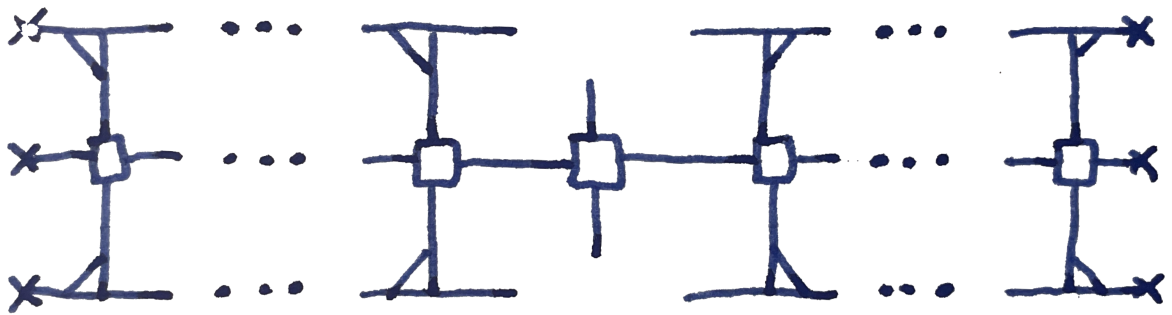


Figure 3.3: DMRG effective Hamiltonian in graphical notation.

state of the effective Hamiltonian as the updated tensor T_i in the ground state of the entire Hamiltonian.

The MPO representation of the Hamiltonian can alternatively be expressed as a sum of, easier to interpret, n -tensor operators. As an example, consider the non-interacting Hubbard Hamiltonian. The hopping terms correspond to identical two-site operators acting on neighbouring sites, while the chemical potential terms are represented by identical one-site operators. In k -space we would only find one-site operators, which would differ for every site. Figure 3.4 demonstrates the effective Hamiltonian in that case.

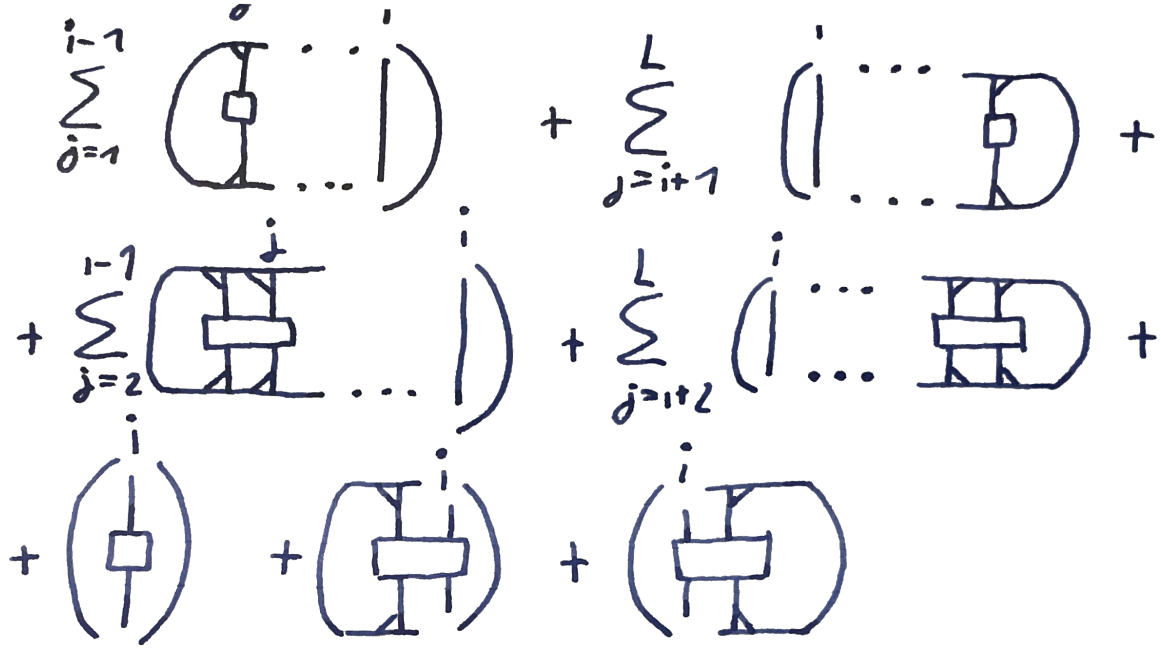


Figure 3.4: Effective Hamiltonian with translation invariant one- and two-site operators.

To find excited states or treat ground state degeneracy with DMRG, one must add another Langrange multiplier term to the cost function (3.1):

$$\langle \psi | \hat{H} | \psi \rangle - \lambda_1 (\langle \psi | \psi \rangle - 1) - \lambda_2 \langle \psi | \phi \rangle - \bar{\lambda}_2 \langle \phi | \psi \rangle$$

where $|\phi\rangle$ is the state and $|\psi\rangle$ should be orthogonal to, for example, the result from the previous DMRG calculation. Again by differentiating with respect to T_i we find the effective vector $|\phi_{eff}\rangle$ sketched in figure 3.5. With this vector one can build the projector:

$$\hat{P} = \mathbb{1} - |\phi_{eff}\rangle \langle \phi_{eff}|$$

and solve the eigenproblem of the effective Hamiltonian (figure 3.2) in the projected space:

$$\hat{P} \hat{H}_{\text{eff}} \hat{P} |\psi_i\rangle = \lambda \hat{P} |\psi_i\rangle$$

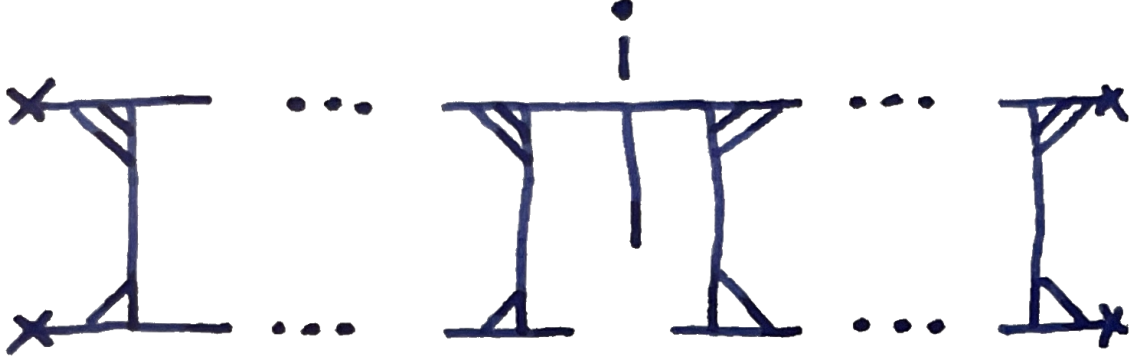


Figure 3.5: Effective Vector $|\phi_{\text{eff}}\rangle$. Where the double triangles denote state $|\phi\rangle$

Generalizing single-site DMRG from an MPS structure to a general TTN structure does not change the essence of the algorithm. The optimization problem still reduces to a one-site effective eigenproblem. However, the traversing pattern needs to be changed to a depth first pattern, where we only start updating the nodes once we reach the first leaf. As we continue with the depth first traversal pattern, every node that is traversed and was not yet updated, will be updated. Traversing here means not just moving from tensor to tensor, but moving the canonical site i . Moving the canonical site i is achieved via QR-decompositions followed by contractions, as in the MPS case. The iteration ends if the canonical site is at the starting point and there are no neighbouring nodes/tensors that have not been updated.

3.3 TDVP

The time-dependent variational principle (TDVP) is a tensor network method, like DMRG, but used to time-evolve a TTN or MPS state. The algorithm is based on the observation that the time derivative of a tensor network lies in the tangent space of that network, i.e. the space spanned by all tensor networks that differ from the original by a single tensor. Thus, if the original network consists of n tensors, its tangent space is spanned by n states, one for each tensor. Crucially, $\hat{H}|\psi\rangle$ does not necessarily lie in this space, and the idea of the TDVP algorithm is to force $\hat{H}|\psi\rangle$ into the tangent space. Figure 3.6 shows the tangent space projector.

Let us briefly argue that the tangent space projector indeed imposes the structure of the derivative onto $\hat{H}|\psi\rangle$. I follow the approach of Jan von Delft, but ignore the notion of kept and discarded spaces in the derivation. I will not show that the tangent space projector is in fact a projector: this is considerably easier with the kept and discarded spaces, and I have

$$\begin{aligned}
 & \sum_{l=1}^L \left(\begin{array}{c} \text{Diagram 1} \end{array} \right) \bigg| \left(\begin{array}{c} \text{Diagram 2} \end{array} \right) \\
 & - \sum_{l=1}^{L-1} \left(\begin{array}{c} \text{Diagram 1} \end{array} \right) \left(\begin{array}{c} \text{Diagram 2} \end{array} \right)
 \end{aligned}$$

The diagrams are Feynman-like diagrams representing tensor contractions. Each diagram consists of two horizontal lines. The top line has a series of vertices (represented by triangles) and the bottom line has a series of vertices (represented by inverted triangles). The vertices are connected by horizontal lines, and there are vertical lines extending from each vertex. The diagrams are separated by a vertical line, and the second diagram is enclosed in large parentheses.

Figure 3.6: Tangent space projector.

nothing to add to von Delft's derivation [7]. We have the QR relationship shown in Figure 3.7 between the site tensors T_i , the left isometries A_i , and the matrices R_i . Differentiating the QR decomposition yields the relation in Figure 3.8 for the derivative of the left isometry A_i . Using contractions and QR decompositions, one can move the matrix R_i and thereby place the canonical site in the desired position. Note that the left-hand side of the equation is then completed by left and right isometries alone. With this relation in hand, we can

The diagram shows a site tensor T_i represented as a horizontal line with a vertical line extending downwards from its center. This is set equal to a left isometry A_i (a horizontal line with a vertical line extending downwards from its left end) followed by a matrix R_i (a circle on a horizontal line).

Figure 3.7: QR decomposition in graphical notation.

The diagram shows the derivative of the left isometry $\dot{A}_i$ followed by R_i is equal to the derivative of the site tensor $\dot{T}_i$ minus the left isometry A_i followed by the derivative of R_i ($\dot{R}_i$).

Figure 3.8: Derivative of the left isometry.

compute the time derivative of the entire MPS. We take the MPS to be in site-canonical form, with the canonical site in the rightmost position L . Figure 3.9 shows the resulting derivative. Comparing its structure to the tangent space projector in Figure 3.6, one immediately sees that the two match exactly. We therefore obtain a differential equation for each tensor T_i and each matrix R_i . We can now use the time-evolution operators provided by these differential equations to evolve the MPS. To see how this works, again assume the MPS is in site-canonical form with the canonical site at L , and start at site 1. We want to find $A_1(t_0 + t)$: applying the time-evolution operator for T_1 gives $T_1(t_0 + t) = A_1(t_0 + t) R_1(t_0 + t)$, from which we read off $A_1(t_0 + t)$. To find $A_2(t_0 + t)$, however, we must time-evolve $T_2(t_0) = R_1(t_0) A_2(t_0)$, which requires evolving the bond matrix R_1 backward in time. The procedure continues in

$$\begin{aligned}
 & \sum_{i=1}^{L-1} \left(\text{Diagram 1} \right) + \left(\text{Diagram 2} \right) \\
 &= \sum_{i=1}^{L-1} \left(\text{Diagram 3} \right) + \left(\text{Diagram 4} \right) \\
 &= \sum_{i=1}^L \left(\text{Diagram 5} \right) - \sum_{i=1}^{L-1} \left(\text{Diagram 6} \right)
 \end{aligned}$$

The diagrams represent tensors in a Matrix Product State (MPS) representation. Diagram 1 shows a chain of tensors from index 1 to L-1, with a derivative operator $\dot{A}_i$ on the i-th tensor. Diagram 2 shows a chain of tensors from index 1 to L, with a derivative operator $\dot{I}_L$ on the L-th tensor. Diagram 3 shows a chain of tensors from index 1 to L-1, with a derivative operator $\dot{A}_i$ on the i-th tensor and a loop labeled R_i on the bond between the i-th and (i+1)-th tensors. Diagram 4 shows a chain of tensors from index 1 to L, with a derivative operator $\dot{I}_L$ on the L-th tensor. Diagram 5 shows a chain of tensors from index 1 to L, with a derivative operator $\dot{I}_i$ on the i-th tensor. Diagram 6 shows a chain of tensors from index 1 to L-1, with a derivative operator $\dot{R}_i$ on the bond between the i-th and (i+1)-th tensors.

Figure 3.9: Derivative of the entire MPS.

the same way until we reach the last tensor $T_L(t_0)$.

Note: The isometries have to be understood as the site-canonical isometries, even in the bond representation. As a consequence, the bond matrices are not necessarily diagonal. Moreover, all references to a QR decomposition imply the reduced (thin) QR decomposition.

3.4 Application: Fast Fourier Transform (FFT)

By comparing the physics convention (left) with the computer-science convention (scipy) on the right in (3.2):

$$\exp(i\omega_k t_n) \stackrel{!}{=} \exp\left(i2\pi k \frac{n}{N}\right) \quad (3.2)$$

one obtains

$$\omega_k := \frac{2\pi k}{t_{\text{end}} - t_{\text{start}}}, \quad t_n := \Delta t \cdot n.$$

3.4.1 Fast Fourier Transform of a Delta Function

We assume the same realization of the delta function that we have been assuming throughout, only we do not take the limit $\eta \rightarrow 0$. This is characterized by a transition from a delta function $\delta(x)$ to a Lorentzian $L(x)$:

$$L(\omega_k - \varepsilon) = \frac{1}{\pi} \frac{\eta}{(\omega_k - \varepsilon)^2 + \eta^2},$$

and we want to calculate the correct normalization constant C of the Fourier transform:

$$C \sum_{k=1}^N \exp\left(i 2\pi k \frac{n}{N}\right) L(\omega_k - \varepsilon)$$

Case: $\Delta\omega \gg \eta$. In this scenario one will only “hit” the delta function if $\varepsilon \in \Omega$ with $\Omega = \{\omega_k \mid k \in [1, N]\}$. The value of the delta function at this position is given by

$$L(\varepsilon - \varepsilon) = \frac{1}{\pi\eta}.$$

Therefore, one has to choose

$$C = \pi\eta.$$

Case: $\Delta\omega \approx \eta$. This case is considered as academic study to create a smoother transition to the next case. In practice, one will either use the previous case, if possible, or approximate the normalization constant by the one in the integration limit. The Fourier transform now requires the normalization factor:

$$C = \frac{\pi}{\sum_k \frac{\eta}{(\omega_k - \varepsilon)^2 + \eta^2}}.$$

Case: $\Delta\omega \ll \eta$. In this limit we assume that

$$\Delta\omega \sum_k \frac{\eta}{(\omega_k - \varepsilon)^2 + \eta^2} \approx \int d\omega \frac{\eta}{(\omega - \varepsilon)^2 + \eta^2} = \pi.$$

And thus we find

$$C = \Delta\omega.$$

This is the most reliable case, and therefore we want $\Delta\omega \ll \eta$. For small η , we must compute the evolution over a long period of time. In the case of imaginary time evolution a small η also requires a long time evolution, to ensure sufficient decay.

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